

Breaking Enrollment Barriers for Refugee Children: Evidence from a Large-Scale Education Support Program

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1. Introduction

Forced displacement has reached historically unprecedented levels. By the end of 2024, 123.2 million people were displaced worldwide, and 40% of them were children (UNHCR 2024). Such exile is often long-term, averaging 17 years, meaning children spend their entire school-age years in these countries of first asylum. Yet enrollment gaps between refugee children and host-country children remain stark: globally, over 46% of school-aged refugee children are out of school, with enrollment rates declining from 67% in primary education to 37% in secondary and just 9% in high education (UNHCR 2025). School enrollment shapes children’s human capital accumulation, long-run economic opportunity, and social cohesion, and access to education is a core right enshrined in international humanitarian law. Thus, ensuring equitable education and integration to the host country is particularly critical to capture the “golden period” of children’s growth.

Despite substantial public and humanitarian investments, these gaps for refugees children remain difficult to bridge. Financial aid programs—such as cash transfers—aim to mitigate economic barriers. However, emerging evidence suggests their impact on schooling is often modest or transient (Hızıroğlu Aygün et al. 2024; Altındağ and O’Connell 2023; Özler et al. 2021). This is largely because enrollment barriers are not purely financial. In practice, factors such as unfamiliarity with host-country systems, language gaps, and school-based marginalization can deter families even when financial constraints are relaxed (Dryden-Peterson 2016). Furthermore, the perceived “purpose” of education is often undermined by legal restrictions that bar refugees from the labor market, diminishing the long-term returns on schooling (Dryden-Peterson et al. 2019). These structural challenges are compounded by low program take-up, as families may remain unaware of available support or find it bureaucratically inaccessible.

Building on these findings, this article examines a different type of intervention: enrollment support programs, that provide tailored information and personal referrals to facilitate school access. A growing body of evidence from general populations demonstrates that informational and navigational interventions can effectively shape schooling decisions by recalibrating perceived returns to education (Jensen 2010), correcting parental misconceptions regarding the school environment (Rogers and Feller 2018), and reducing administrative frictions (Bettinger et al. 2012). However, the transferability of these findings to refugee contexts—where barriers are qualitatively more complex—is not yet established. The preceding discussion suggests reasons for skepticism: if refugee families confront compounding obstacles that are simultaneously legal, linguistic, economic, and psychosocial in nature, then reducing informational and navigational frictions alone may be insufficient to move families toward enrollment. At the same time, the very identification of which margin is binding for a given family is itself uncertain—and may vary substantially across households facing different combinations of obstacles. Whether enrollment support can generate meaningful gains in this environment is therefore a genuinely open empirical question, and one with direct policy implications given the scale of displacement and the limited effectiveness of costlier alternatives.

Enrollment support programs operate through frontline workers who proactively contact families, identify out-of-school children, and provide hands-on assistance navigating the enrollment process. This direct, case-by-case mode of delivery is also what makes such programs operationally constrained. On the supply side, each outreach contact requires substantial field-worker time; each case must conform to local education policies and school-level practices; and effective outreach demands significant investment in training workers to navigate complex systems and build trust with families. On the demand side, the population of refugee children in need of support typically far exceeds campaign capacity and continues to grow, creating a critical operational tension between reach and effectiveness of the program.

(motivate fairness question here)

Motivated by these observations, we seek to answer two questions. *First*, does enrollment support causally improve enrollment and short-term retention for out-of-school refugee children, and how does its impact vary across demographic and contextual factors? *Second*, given the limited program capacity and its heterogeneous impact on refugee families, how should the program allocate its scarce outreach resources—deciding who to support and when—to maximize impact while respecting principles of fairness?

We study these questions by evaluating the largest enrollment support program for refugee children in Turkey. Turkey hosts the world’s largest refugee population, with over 3.7 million registered Syrian refugees alone (UNHCR 2024), making it a critical setting for understanding how outreach can reduce enrollment barriers at scale. We collaborate with the Association for

Solidarity with Asylum Seekers and Migrants (ASAM), the largest non-profit organizations serving refugees and asylum seekers in Turkey. ASAM has over 30 years of experience in its area and operates 29 offices across major cities. On January 1, 2020, ASAM launched the Support for School Enrollment (SSE) program designed to address informational and procedural barriers to schooling for refugee children. SSE began in eight cities and later expanded to ten. By March 2025, the program had covered over 330,000 refugee children. SSE staff proactively contact families, conduct education assessments to determine schooling status and barriers, and—in collaboration with the local education ministry—provide hands-on assistance including form filling, translation, and accompaniment to help families register children at nearby schools. Crucially, the number of registered out-of-school children substantially exceeds the outreach team’s capacity in any given enrollment season, creating operational tension between reach and depth of support.

We study these questions in the context of Support for School Enrollment (SSE) program, a large scale enrollment support program for refugee children in Turkey. Turkey hosts over 3.7 million registered Syrian refugees (UNHCR 2024), making it a critical setting for understanding whether outreach can reduce enrollment barriers at scale. The SSE program was launched in January 2020 by our collaborator, the Association for Solidarity with Asylum Seekers and Migrants (ASAM), Turkey’s largest refugee-serving non-profit. It had covered over 330,000 children across ten cities by March 2025. SSE staff proactively contact families, assess schooling status and barriers, and—in coordination with the local education ministry—provide hands-on assistance including form filling, translation, and accompaniment to schools.

We exploit a distinctive feature of the program’s operational design to identify its causal effect. Within each city, children enter the assessment pipeline in a computer-generated random order. Whether a child receives support before the annual outcome evaluation depends primarily on the child’s position in this queue and available staff capacity, not on the child’s characteristics or expected outcomes—consistent with UNHCR’s principle of impartiality (UNHCR 2015). Children who receive support before the fall enrollment deadline are considered treated; those whose support is delayed beyond the deadline serve as controls. Using the organization’s administrative case-management database spanning 2020–2025 and covering 177,116 out-of-school children ages 5–17, we estimate treatment effects using causal machine learning methods (Chernozhukov et al. 2018a, Athey et al. 2019) to characterize treatment-effect heterogeneity at the individual level.

(continue from here.)

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the background and data. Section 4 develops our causal identification and estimation strategy. Section 5 presents the results including heterogeneous treatment effect and robustness check. Section 6 formulates and solves the fair policy targeting problem for a better program design. Section 7 concludes.

2. Related Literature

Humanitarian Operations for Refugee Services. Operations management research on refugee services has focused predominantly on resettlement—the placement of refugees in a third country—developing data-driven optimization models for matching and dynamic allocation (Bansak et al. 2018, Ahani et al. 2021, Delacrétaz et al. 2023, Ahani et al. 2023, Bansak et al. 2024). However, resettlement is available to less than 1% of the refugee population (Dryden-Peterson et al. 2019); for the vast majority, exile in the host country lasts 10 to 25 years. As a result, since 2012, the priority of the UNHCR Global Education Strategy has shifted to *post-settlement integration* (United Nations High Commissioner for Refugees 2012, 2019). We complete this research gap by studying integration strategies with a focus on education. This integration phase poses a distinct set of operational challenges: identifying families in need, delivering individualized assistance at scale, and allocating scarce caseworker time across a large population whose needs are heterogeneous, evolving, and seasonal. We provide the first large-scale empirical study of a social integration program, combining causal evaluation with operational design. Our focus is education for refugee children because of its critical importance for long-term integration, but the program’s design and mechanisms exemplify the broader challenge of delivering individualized support at scale.

Policy Interventions for Refugee Enrollment Gaps. The most prominent policy response to improve refugee education has been to ease the financial burden of schooling. Turkey’s Emergency Social Safety Net (ESSN) provides unconditional cash transfers to vulnerable refugee households, while the Conditional Cash Transfer for Education (CCTE) ties payments to children’s regular school attendance. The ESSN alone has disbursed over €3.8 billion since 2016. Despite substantial investment, the evidence on enrollment effects is mixed. Hızıroğlu Aygün et al. (2024) find that the CCTE program increased attendance among children already enrolled but had limited effects on bringing out-of-school children into the system. Altındağ and O’Connell (2023) document only transient enrollment impacts from unconditional cash transfers to Syrian refugees in Lebanon. Özler et al. (2021) report that the ESSN in Turkey had no significant direct effect on school enrollment despite meaningful reductions in poverty and food insecurity. This pattern also holds for general population. Across a range of developing-country contexts, researchers have found that removing cost barriers is often necessary but not sufficient for increasing educational participation (see Glewwe and Muralidharan 2016 for a review). Our work consolidates a new policy tool, enrollment support program, which is increasingly adopted by humanitarian organizations but has not been rigorously evaluated in the literature of refugee education. We demonstrate that the informational and procedural interventions in the program can significantly increase enrollment probability and short-term retention for refugee children.

Informational and Navigational Assistance in Education. A growing literature demonstrates that informational and navigational assistance can meaningfully shift education decisions—even when financial constraints remain unchanged—by correcting beliefs about returns to schooling (Jensen 2010), reducing misperceptions about attendance (Rogers and Feller 2018), and increasing awareness of financial aid (Dinkelman and Martínez A. 2014). The most closely related study is the H&R Block FAFSA experiment (Bettinger et al. 2012), which marked a key shift from information provision to *navigational assistance*: low-income families who received hands-on help completing a financial-aid application experienced an 8 percentage-point increase in college completion, whereas families receiving only information showed no improvement. The experiment established that the binding friction was administrative complexity, not ignorance—and that personal assistance could resolve what information alone could not. Our setting extends this paradigm to a qualitatively different context. Addressing the barriers faced with refugee children requires sustained case management rather than a one-time intervention, making the program operationally intensive and capacity-constrained in ways that prior informational experiments were not. The scale of our program also allows us to estimate average effects with precision and to uncover rich heterogeneity informing targeted program design.

Causal Machine Learning and Fair Policy Targeting. We employ causal machine learning, including double/debiased machine learning (Chernozhukov et al. 2018a) and generalized random forests (Athey et al. 2019), to estimate heterogeneous treatment effects at the individual child level. These methods have been increasingly applied to operational problems, including tax-induced inequalities in the sharing economy (Cui and Davis 2022), last-mile delivery targeting (Lu et al. 2023), and emergency-department bed assignment (Liu et al. 2025). Such individualized estimates naturally pair with the policy learning framework of Athey and Wager (2021), which uses CATE estimates to derive optimal treatment assignment rules under capacity constraints.

However, naively prioritizing children with the highest estimated treatment effects can systematically favor certain demographic groups, raising fairness concerns that are especially acute in humanitarian settings. (some more literature here). We contribute to the fair policy targeting literature by proposing a new formulation representing UNHCR’s principle of impartiality (UNHCR 2015), which endogenously aligns efficiency with equity since the most vulnerable children have the largest scope for gain. We further integrate the program’s two-stage service structure and seasonal capacity constraints into the nonmaleficence framework of Viviano and Bradic (2024), bridging the gap between individualized policy learning and the practical constraints of humanitarian program design.

3. Background

3.1. Overview of the Program

We collaborate with the Association for Solidarity with Asylum Seekers and Migrants (ASAM), the largest non-governmental organizations serving refugees and asylum seekers in Turkey. Founded in 1995, ASAM operates 29 offices across major Turkish cities and maintains an administrative case-management database that tracks all registered refugee household members and their interactions with ASAM services. Beyond responding to walk-in requests, ASAM proactively maintains contact with registered families through phone calls and home visits and conducts periodic protection assessments to identify needs and deliver services in a timely manner.

Through decades of frontline work with refugee families, ASAM staff observed a persistent pattern: despite improvements in housing and employment stability for many Syrian refugee households, a large share of school-aged children remained out of school. ASAM’s education assessments, a structured evaluations conducted with each out-of-school child reached by the program, repeatedly surface procedural and informational obstacles, rather than financial hardship, as the most widespread barriers to enrollment. Across 110,067 first-contact assessments conducted in the eight cities we study from 2021–2023, 40.5% of children cite school-access barriers — difficulty completing re-registration, lack of knowledge of a recognized enrollment channel, or being refused at the school gate. By comparison, financial hardship is identified as a primary barrier in only 9.1% of cases, 9.4% of children refuse school themselves (often tied to bullying and peer influence). Figure 1 reports the prevalence of each barrier at first contact, split by schooling status (drop-out vs. never-attended); the within-status breakdown across the three out-of-school branches is reported in Figure EC.3.

On January 1, 2020, ASAM launched the Support for School Enrollment (SSE) program, funded by UNICEF and the European Union in collaboration with the Turkish Ministry of National Education (MoNE).¹ SSE provides individualized case-based support to help reintegrate out-of-school refugee children into the formal education system. The program began in eight cities — Gaziantep, Istanbul, Adana, Şanlıurfa, İzmir, Bursa, Mersin, and Konya — covering the largest concentrations of refugee families. In mid 2024, Hatay and Kahramanmaraş joined following post-earthquake reconstruction efforts. Each city maintains a dedicated SSE team with staffing levels scaled to the local registered population. By March 2025, SSE had reached over 330,000 refugee children across these ten cities. Figure 2 summarizes the program’s operational timeline and highlights the 2021–2023 cohort window used in our analysis, which avoids both the COVID-19 disruption of the 2020 cohort and the confounding effect of our own research collaboration that began in 2024.

¹ For UNICEF’s public-facing description of SSE, see <https://www.unicef.org/turkiye/en/support-school-enrolment-sse-programme>; the program’s history, current geographic footprint, and partnership architecture are summarized in greater detail in Appendix EC.1, with field vignettes drawn from the UNICEF Türkiye photo-essay series in Appendix EC.1.5.

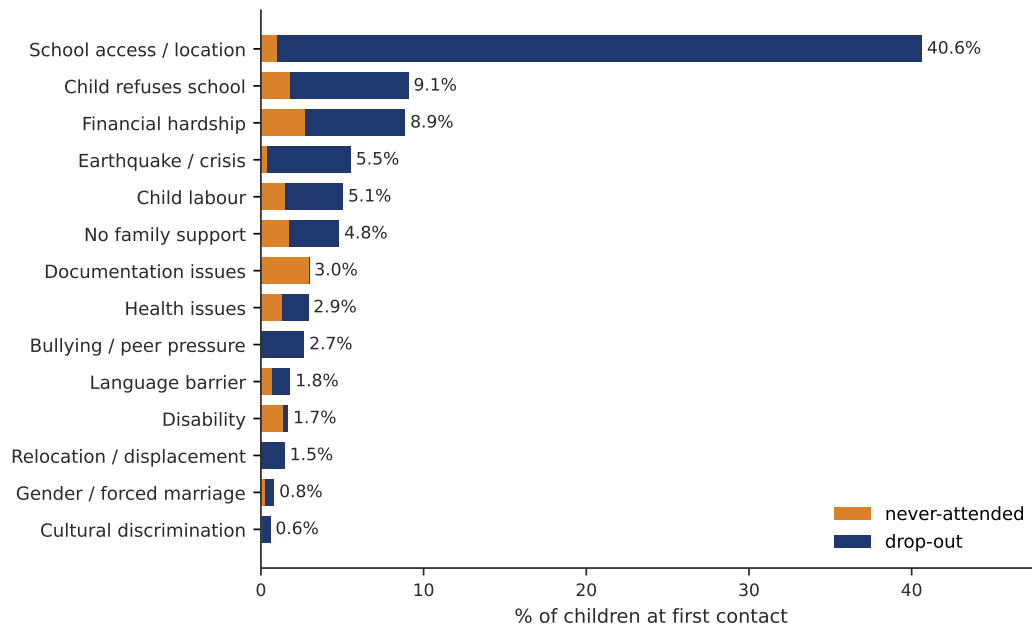


Figure 1 Schooling barriers at first contact, 2021–2023 analysis sample ($N = 110,067$ first-entry out-of-school children). Each row reports the share of children who cite a given barrier on the first-contact assessment, split by current schooling status: drop-out (navy) for children with prior school experience who have left, never-attended (amber) for children who have never enrolled. Bars are ordered by total prevalence and the labels at right report the combined share. Informational and procedural barriers (school access / location, child refuses school, financial hardship) dominate; the irregular-attendance branch and the “not of school age” category are excluded from this view because they are not SSE-actionable at first contact.

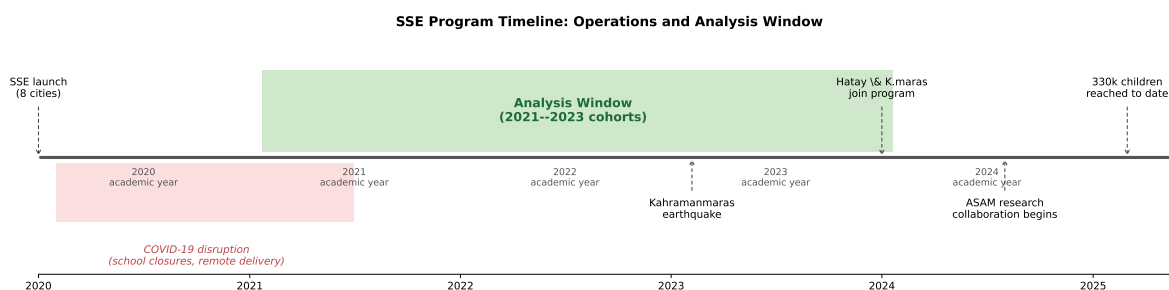


Figure 2 SSE program timeline (Jan 2020 – Mar 2025). The shaded green band marks the 2021–2023 cohort window used for estimation; the red band marks the COVID-19 disruption that compromised the 2020 cohort.

We include more details of ASAM and the SSE program including service environment and field outreach in Appendix EC.1. We attach three short beneficiary vignettes drawn from UNICEF’s photo-essay series (Appendix EC.1.5) for the SSE program.

3.2. Children’s Trajectory through the Program

The SSE program operates through a structured pipeline that moves each child from initial contact through assessment, treatment delivery, and follow-up. Figure 3 shows a child common trajectory in the program.

Entry into the pipeline. ASAM maintains a registry of refugee families and their household members with their residence and contact information. Within each city, the SSE team contacts families for education-specific outreach following the order of a computer-generated list. The list is generated from ASAM’s administrative registry, which records demographic and contact information. At the time the queue is constructed, the detailed information about each child’s schooling status, barriers, and education background that the program collects through its assessments has not yet been obtained. The sequencing therefore does not reflect caseworker judgment about which children are more likely to benefit or more urgently in need of support. Because the number of registered out-of-school children substantially exceeds the team’s contact capacity in any given period, a child’s position in this queue together with available staff capacity largely determines when the child enters the program pipeline.

Education assessment. Upon reaching a family, SSE field workers conduct a structured education assessment. The assessment serves two purposes: to verify the child’s current schooling status and to identify the specific barriers preventing enrollment. Field workers administer a standardized questionnaire through direct contact with the family (typically the parents, sometimes the child if the child is over 14 years old). The questionnaire first asks about the child’s education history and current enrollment and attendance status. Then, conditional on the current school status, they further investigate the barriers to school access for never-enrolled children, reasons for dropout among prior attendees, and barriers to regular attendance for enrolled children. Finally, the questionnaire captures information about existing support and preparation (e.g., scholarships, language courses, material aid). The questionnaire locates each child along the school enrollment pipeline, enabling caseworkers to identify the appropriate intervention for each child. The barrier sections are particularly detailed, employing a multi-select inventory spanning economic factors, documentation issues, gender-based barriers, language difficulties, school-side constraints, safety concerns, displacement-related disruptions and blocks for self-filling.

Treatment delivery. Conditional on the needs and barriers identified in the assessment, the SSE team provides support combining two complementary components. First, field workers provide *information and counseling* to parents: explaining the importance and feasibility of re-enrollment, the practical steps required, relevant legal protections for refugee children’s access to education, and available support services. Second, the team facilitates *external referrals and school placement*

for children: connecting families to specific schools, assisting with required paperwork and documentation, providing translation support, and in some cases physically accompanying families to schools to complete the registration process. These treatments are coordinated with local MoNE offices to ensure that referrals align with school capacity and enrollment procedures. Each treatment is recorded as a discrete event in the case-management system with a timestamp, and a child may receive multiple treatments over time as circumstances evolve.

Follow-up and outcome verification. Following treatment delivery, the program monitors each child’s schooling trajectory. SSE staff verify enrollment status through direct contact with families and cross-checking with school records. The checking of school status and case outcomes are collectively conducted after the fall semester every year, to align with the annual enrollment cycle and to allow sufficient time for the program’s interventions to take effect. Once a child successfully enrolls, the program provides stationery kits to encourage continued attendance. The program continues to track attendance and engagement through subsequent semesters, maintaining contact with the family to identify and address any emerging risks of disengagement. This longitudinal follow-up continues until the child completes basic education.

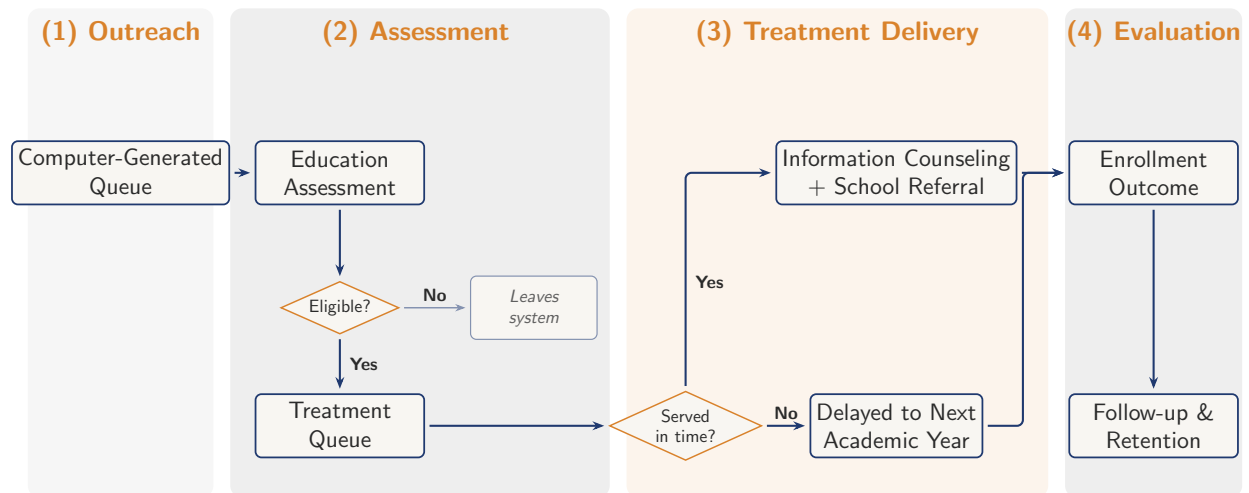


Figure 3 SSE program operation flow. A child enters the pipeline through a computer-generated outreach queue (Stage 1), receives a structured education assessment that determines eligibility (Stage 2), waits in the city-level treatment queue, and is either served before the end-of-fall-semester cutoff (treatment group) or delayed to the next academic year (control group) (Stage 3). Both treated and delayed children are followed up to verify the fall-semester enrollment outcome and tracked into the next academic year for retention (Stage 4).

3.3. The Cohort Structure of the Program

The SSE program has operated continuously since January 2020, spanning five academic years. The program operation is strongly shaped by Turkey’s academic calendar. Most importantly, school enrollment is concentrated in the fall semester as the start of an academic year². The organization responds accordingly, allocating additional capacity to SSE during the enrollment season from August through October. Appendix Figure EC.11 illustrates this seasonality, plotting the monthly volume of education assessments and treatment deliveries from 2020 to 2024. The recurring pattern—sharp peaks during enrollment season, reduced activity during off-season months—is evident in each program year. Additionally, within each city, one SSE team was built and dispatched for operations of SSE program. Each maintains its own queue of registered out-of-school children, its own staffing allocation, and its own capacity constraint. Except for transferring the cases to nearby cities office if the children or family is moved, the SSE team in each city work independently³. The annual rhythm, combined with the program’s city-level operational structure, creates a natural cohort design. For each cohort we have a well-defined quasi-experiment conducted. Children within the same city-year cohort share similar institutional conditions: the same local policy environment, school availability, team capacity, and queue for the treatment. Across cohorts, conditions vary in staffing, the local refugee population, and exposure to aggregate shocks⁴.

We define a *city-year cohort* for city c and academic year t as the set of children living in city c who receive their first education assessment during academic year t . The cohort structure naturally aligns with the Turkish academic calendar: it spans from the end of the previous year’s fall semester (approximately mid-January of year t) to the end of the current year’s fall semester (approximately mid-January of year $t + 1$). The *treatment cutoff* coincides with the end of this assessment window which marks the point after which support delivery can no longer facilitate enrollment in the current academic cycle. The *outcome observation window* extends from the treatment cutoff to the end of the following fall semester (approximately January of year $t + 2$), or until the child’s next out-of-school treatment episode, whichever comes first. This extended window allows sufficient time to observe both enrollment outcomes and retention through the subsequent academic year, while censoring at the next treatment episode preserves the clean comparison between timely and delayed support. Figure 4 illustrates this structure across the five academic years. The cohorts consist eight cities operating continuously from 2021-2023 academic years are the most valuable

² According to Turkey’s law, mid-semester enrollment is technically feasible for refugee children. However, families are reluctant to re-enter during the middle of the academic year. (more evidence needed)

³ We verify this empirically: 96.5% of caseworker events occur within each worker’s primary city, and 95.3% of caseworkers have at least 90% of their recorded events in a single city.

⁴ Cohort structure could absorb disruptions such as the COVID-19 pandemic from 2020 to 2021 and the February 2023 earthquake affecting southeastern Turkey.

for clean causal identification, because in 2020 the program is still under preparation and cohorts were heavily disrupted by the COVID-19 pandemic, which caused school closures and shifted the program’s operations to remote delivery, while in 2024 we started collaboration with ASAM and influences has been observed in the program operation in 2024 fall.

Within each city-year cohort, the program faces a fundamental capacity constraint: the number of out-of-school children awaiting support substantially exceeds what the local SSE team can process before the enrollment deadline. As a result, only a fraction of assessed children receive action delivery in any given cycle. The primary allocation rule is first-come, first-served: because families are contacted following the computer-generated list, children reached earlier in the queue are more likely to receive timely support. Among treated children, the rank correlation between queue position and service date is 0.86, confirming that caseworkers process cases in the prescribed order rather than selectively advancing particular cases.⁵ Alongside this sequencing rule, the organization makes deliberate capacity allocation decisions reflecting its assessment of need and urgency. Children who previously attended school in Turkey but dropped out are prioritized over those who have never enrolled, as they are at greater risk of permanent disengagement and face a narrowing window for re-entry. Children at school transition ages (around ages 6 and 10, corresponding to primary and middle school entry) are prioritized because missing these critical enrollment windows creates compounding disadvantages as the curriculum gap widens. Children from single-mother households receive greater outreach effort because the organization identifies these families as more vulnerable⁶. These priorities are consistent with UNHCR’s principle of impartiality, which calls for allocating resources based on assessed need rather than treating all cases identically. Impartiality is distinct from strict equality of treatment.⁷

The program’s administrative case-management database records the complete history of interactions for each child: education assessments, action deliveries, follow-up contacts, and verified schooling outcomes, all with timestamps. At the time of assessment, the system also captures a rich set of child and household characteristics including demographics, education history, language proficiency, family structure, economic circumstances, and identified barriers to schooling. These records, spanning five years and ten cities, constitute the data foundation for the empirical analysis that follows.

⁵ We measure this using Kendall’s τ between assessment date and action delivery date within each city-year cohort. Table EC.8 reports the correlation for each cohort; the median across cohorts is 0.86.

⁶ Approximately 50% of children in our sample live in single-parent households. Among these, single-mother households are more common than single-father households, consistent with patterns of family separation during displacement.

⁷ Impartiality means that humanitarian action is based solely on need, with priority given to the most urgent cases irrespective of factors such as race, nationality, gender, religious belief, political opinion, or class. UNHCR’s humanitarian principles, humanity, impartiality, neutrality, and independence, guide all emergency preparedness and operational responses. They are rooted in international law and UN resolutions, and are the foundation of its mission to protect refugees and displaced persons.

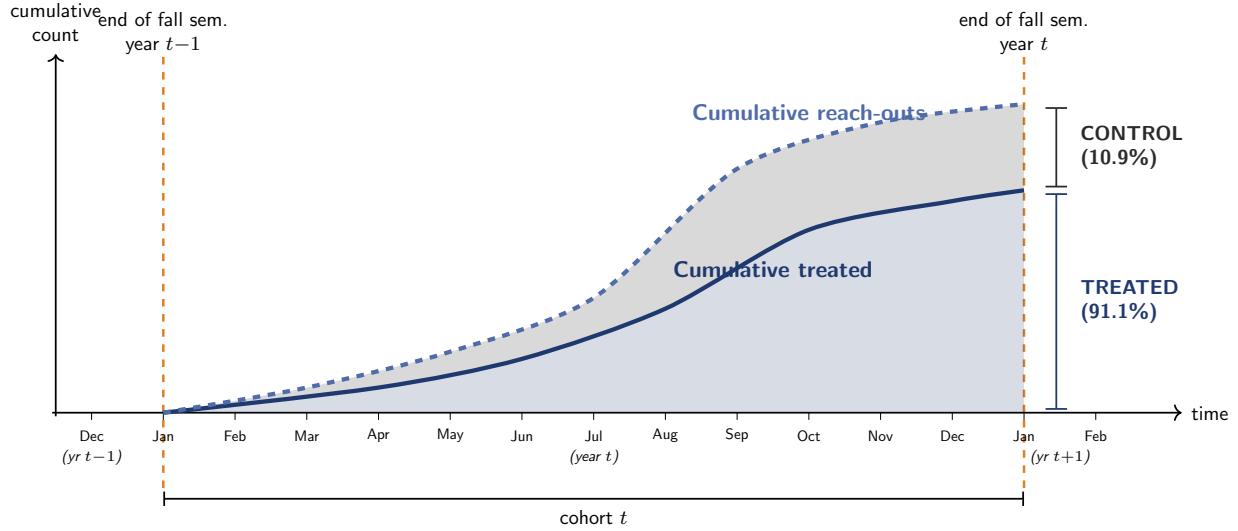


Figure 4 Cohort structure and treatment allocation within a single city-year cohort t . The cohort window runs between two consecutive end-of-fall-semester cutoffs (vertical dashed lines). The dashed upper curve plots cumulative reach-outs by the SSE team — the total set of children the team contacts during the window. The solid lower curve plots cumulative treated — children whose first counseling and referral session is delivered in time for the fall enrollment cycle; the FIFO service rate is slower than the reach-out rate, so a queue of contacted-but-not-yet-served children builds up between the two curves. At the cohort cutoff the two regions realise the experimental groups: the solid-navy region below the treated curve is the treatment group (91.1% pooled across the 24 city-year cohorts), and the grey region between the two curves is the control group (residual queue, 8.9%), as bracketed on the right.

4. Identification and Estimation Strategy

4.1. Identification Strategy

The central empirical challenge is to isolate the causal effect of receiving enrollment support from the many confounding factors that jointly influence both service delivery and schooling outcomes. A naive comparison of supported and unsupported children would therefore conflate the program’s causal impact with systematic selection into treatment. Our identification strategy exploits the SSE program’s operational design described in section 3. First, the treatment assignment generally follows the exogenous reaching out sequence determined by computer generated random list, so the earlier arrivals receive treatment while the later ones do not. Second, we have clear selection rules based on observable characteristics which are all recorded at the time of assessment. Third, the program’s annual rhythm creates a natural cohort structure, with each city-year cohort constituting a self-contained unit with its own intake, capacity constraint, treatment delivery, and outcome evaluation. Conditional on these observables, remaining variation in treatment timing is driven by queue position and residual staff capacity: factors that are, by design, unrelated to the child’s potential enrollment outcome. The identifying assumption is therefore conditional independence,

that treatment assignment is as good as random after conditioning on the observed covariates and fixed effects. We show the unconditional and conditional covariates balance in Table EC.14.

We define treatment at the child level within each cohort: The treatment group consists of children who are supported before the end of fall semester of the year entering the program, which is the treatment cutoff. This timing ensures that the support can directly facilitate enrollment in the current academic cycle. While the control group includes children who were assessed within the same cycle but whose action delivery was delayed beyond the enrollment window, either receiving support later in the academic year, in a subsequent cycle, or not at all⁸. The estimand is the causal effect of *timely* enrollment support relative to the counterfactual of delayed or absent support, where people need to enroll by themselves (the natural trend of enrollment). This is the policy-relevant quantity: given binding capacity constraints, the operational question facing the program is not whether to serve a child eventually, but when to prioritize that child in the queue.

The treatment allocation repeats itself in different cities and different years. As described in Section 3.3, the program’s annual rhythm creates a natural cohort structure: each city at each enrollment cycle constitutes a self-contained unit with its own intake, capacity constraint, treatment delivery, and outcome evaluation. Within each cohort, the assignment mechanism remains the same. This structure yields a series of independent quasi-experiments, analogous to the repeated lottery designs in Angrist et al. (2002), each providing a separate estimate of the program’s causal effect under different contextual conditions.

Table 1 reports the resulting sample composition. Restricting to each child’s first entry into the SSE pipeline yields an analysis sample of 110,067 children across the three enrollment cycles, of whom 100,223 (91.1%) received timely support before the treatment cutoff and 9,844 (8.9%) constitute the control group. Pooled across cohorts, 67.1% of the treated children enrolled and 45.0% were retained; among the control group, these rates are 4.7% and 1.4%, respectively, yielding raw percentage-point differences of +62.5 for enrollment and +43.6 for retention. (do I need to specifies the variations across cohorts?)

We track two binary outcomes over the observation window. *Enrollment* (Y^E) equals one if the child has a verified school enrollment within the window. The second outcome measures whether that enrollment *persists*: Y^R equals one if the child is verified still attending at the close of the window. Because a child can be retained only if first enrolled ($Y_i^R = 1 \Rightarrow Y_i^E = 1$) and we measure Y^R *unconditionally* over the full sample — setting $Y_i^R = 0$ for every child who never enrolls —

⁸ Our analysis restricts attention to each child’s first entry into the SSE pipeline. Subsequent entries are excluded because they are triggered by outcomes of earlier interventions (e.g., re-dropout after initial re-enrollment), introducing post-treatment selection that would compromise comparability.

Y^R is the *joint* event “enrolls *and* stays.” We therefore call Y^R *durable enrollment* (enroll-and-retain) and read it as the part of the enrollment gain that is still in place a semester later, not as a conceptually separate retention channel; for brevity, figures and tables retain the short label *retention*. A zero on Y^R pools two distinct situations — a child who never enrolled and a child who enrolled but later dropped out — and the second cell is well populated: 33% of enrollees are not retained, so Y^E and Y^R are far from collinear (child-level correlation 0.66). We deliberately do *not* define retention *conditional* on enrollment. Doing so would condition on a post-treatment variable and compare non-comparable populations — the control group enrolls at only 4.7%, so the enrolled-control cell is nearly empty — and the within-enrollee staying margin is consequently not separately identified in these data. Durable enrollment is the causally clean object; we return to how its heterogeneity relates to the enrollment surface in subsection 5.2.

Table 1 Sample Statistics by City-Year Cohort. Each row gives the cohort size, share of children who received timely enrollment support (treated), and the enrollment and retention rates by treatment status, along with their raw percentage-point difference. The final row pools all 24 cohorts and reports whole-sample totals (for N) and weighted means (for all percentage columns).

Cohort	N	% Treated	Treated		Control		Difference	
			Enrolled	Retained	Enrolled	Retained	Enrolled	Retained
Gaziantep 2021	7,118	94.4%	63.0%	46.7%	18.7%	6.5%	+44.3	+40.2
Istanbul 2021	5,451	89.5%	68.5%	54.4%	27.3%	10.9%	+41.1	+43.6
Izmir 2021	7,208	79.9%	92.2%	75.1%	6.5%	2.6%	+85.7	+72.5
Adana 2021	4,183	90.3%	81.1%	63.9%	9.1%	2.5%	+72.1	+61.4
Sanliurfa 2021	3,061	93.2%	73.3%	53.5%	17.3%	6.7%	+55.9	+46.8
Mersin 2021	3,214	96.1%	80.5%	66.3%	14.3%	5.6%	+66.2	+60.8
Bursa 2021	2,733	93.1%	83.2%	71.6%	14.8%	7.9%	+68.4	+63.6
Konya 2021	1,890	94.0%	65.1%	49.3%	14.0%	4.4%	+51.1	+44.9
Gaziantep 2022	9,172	89.0%	53.6%	30.7%	10.2%	2.6%	+43.4	+28.2
Istanbul 2022	5,808	80.7%	62.8%	41.6%	18.3%	7.5%	+44.5	+34.1
Izmir 2022	4,634	74.1%	78.9%	51.8%	7.2%	2.9%	+71.7	+48.9
Adana 2022	4,708	95.1%	75.5%	54.1%	9.9%	2.6%	+65.6	+51.5
Sanliurfa 2022	4,452	94.4%	76.5%	43.9%	18.5%	5.6%	+57.9	+38.2
Mersin 2022	4,419	94.7%	72.7%	51.3%	14.7%	9.9%	+58.0	+41.4
Bursa 2022	2,836	91.4%	73.4%	50.4%	18.5%	9.5%	+54.9	+41.0
Konya 2022	3,718	95.2%	66.1%	33.5%	30.2%	13.4%	+36.0	+20.1
Gaziantep 2023	5,780	87.0%	71.1%	36.6%	27.5%	9.0%	+43.7	+27.6
Istanbul 2023	5,159	80.1%	69.5%	46.6%	15.7%	5.4%	+53.9	+41.2
Izmir 2023	4,198	66.2%	85.8%	51.8%	22.8%	6.1%	+63.0	+45.7
Adana 2023	3,637	92.8%	69.7%	37.2%	10.3%	2.7%	+59.4	+34.5
Sanliurfa 2023	3,297	96.4%	86.9%	68.3%	21.7%	8.3%	+65.2	+60.0
Mersin 2023	3,692	93.8%	72.9%	53.8%	17.5%	6.1%	+55.4	+47.7
Bursa 2023	4,024	88.6%	75.2%	51.9%	23.7%	6.7%	+51.5	+45.2
Konya 2023	3,289	92.6%	64.3%	31.4%	30.2%	13.2%	+34.1	+18.2
Overall	107,681	88.4%	72.2%	49.6%	16.3%	5.7%	+56.0	+43.9

Two features of the design require explicit specification choices. First, children assessed earlier in the cycle mechanically have more time both to receive action delivery and to experience schooling outcomes before the evaluation window closes. We address this by including entry-month fixed effects, which restrict comparisons to children assessed in the same calendar month. Within entry-month cells, variation in treatment status is driven by queue position and residual staff capacity — the quasi-random component of the assignment mechanism. Second, children within the same family may share unobserved shocks, parental preferences, and information, and may respond jointly to outreach. To account for within-family correlation in unobservables, we cluster standard errors at the family level.

The identification strategy thus rests on conditional independence: after absorbing cohort-level heterogeneity through city-year fixed effects, time-at-risk through entry-month fixed effects, and priority-based selection through the observed covariate vector, residual variation in treatment timing is plausibly exogenous. This conditional independence assumption — rather than unconditional randomization — is precisely what the double/debiased machine learning framework is designed to exploit. The flexible first stage can absorb nonlinear relationships between the priority-determining covariates and both treatment and outcomes without requiring correct parametric specification, while cross-fitting prevents overfitting. Moreover, the average treatment effect alone is insufficient for the program design question: with capacity binding in every enrollment cycle, the operational question is not merely whether the program works, but for whom it works most and how to allocate scarce resources. We develop the estimation strategy in the next subsection.

4.2. Estimation Strategy

The paper’s research objectives extend beyond the average treatment effect. We pursue three progressive goals: first, to estimate the average causal effect of the program on enrollment and retention; second, to characterize how this effect varies across children as a function of their demographics, household circumstances, and identified barriers; and third, to use individual-level treatment effect estimates as inputs to the policy targeting problem developed in section 6. The first goal alone could be addressed by conventional regression. The second and third require an estimator flexible enough to capture nonlinear interactions across a high-dimensional covariate space without pre-specifying the functional form of heterogeneity. We develop a unified estimation framework that addresses all three goals within a single coherent structure.

Our priority is the heterogeneous treatment effect surface itself rather than a single average effect. The program operates under a binding capacity constraint and the policy question is allocation, not aggregate effectiveness; the right object is therefore the per-child CATE function $\tau(x)$ on the full covariate vector, not a few hand-picked moderators. We adopt the causal forest as the *primary*

second-stage estimator on top of a shared first-stage residualization, and use the DML average-treatment-effect estimator (DML-ATE) as a single-line backup point estimate. We deliberately do not pre-specify moderator interactions for confirmatory linear inference: with the high-dimensional risk-needs and barrier covariates introduced in section EC.6 (over 180 controls including ~ 64 sparse text-encoded dummies), pre-specifying which interaction terms to test would be arbitrary, and any reasonable correction for multiple comparisons would deplete the formal power that pre-specification is meant to deliver. The data-driven CATE surface, together with the systematic 1D / 2D / outcome-contrast exploration in subsection 5.2, is more honest about what the estimation can speak to.

4.2.1. The R-Learner Framework All estimators in this paper are unified under the R-learner framework (Nie and Wager 2021), which builds on Robinson’s (1988) partially linear model. The core idea is to remove the influence of confounding covariates from both the outcome and the treatment through flexible first-stage regressions, then estimate the treatment effect function from the resulting residuals. Specifically, let $m(X_i, C_i) = \mathbb{E}[D_i | X_i, C_i]$ denote the conditional treatment probability and $g(X_i, C_i) = \mathbb{E}[Y_i | X_i, C_i]$ denote the conditional outcome expectation, where X_i is the vector of individual covariates and C_i indexes the city-year cohort. Define residuals $\tilde{D}_i = D_i - \hat{m}(X_i, C_i)$ and $\tilde{Y}_i = Y_i - \hat{g}(X_i, C_i)$. The R-learner estimates the treatment effect function $\tau(\cdot)$ by minimizing the R-loss:

$$\hat{\tau} = \arg \min_{\tau} \sum_{i=1}^n \left[\tilde{Y}_i - \tau(X_i) \cdot \tilde{D}_i \right]^2 \quad (1)$$

Two estimators in this paper differ only in the function class imposed on τ . When $\tau(X_i)$ is estimated nonparametrically via a causal forest (Athey et al. 2019), minimization yields individual-level conditional average treatment effect (CATE) estimates. This is our primary object of interest. When $\tau(X_i) = \theta_0$ (a constant), the same R-loss reduces to the double/debiased machine learning (DML) estimate of the average treatment effect (Chernozhukov et al. 2018a), which we report as a backup sanity check. The shared first-stage residualization is identical for both estimators, so the comparison is purely about the function class imposed on the second-stage CATE.

4.2.2. Shared First-Stage Residualization The first stage, shared by all downstream estimators, removes confounding variation from both the outcome and the treatment. We estimate the nuisance functions $m(X_i, C_i)$ and $g(X_i, C_i)$ using gradient boosting, where the conditioning set includes the full vector of individual covariates X_i — child demographics, household characteristics, identified barriers to schooling, and entry-month indicators — alongside city-year cohort indicators C_i . Entry-month indicators serve a dual role in the estimation: within the nuisance functions, they

absorb the time-at-risk confound identified in subsection 4.1, ensuring that residuals reflect within-entry-month variation in treatment timing; in subsequent stages, they also enter the heterogeneity estimation, allowing the treatment effect to vary with assessment timing.

City-year cohort indicators enter *only* the nuisance stage and are excluded from the covariate set in the heterogeneity estimation. This design choice reflects the structure of the policy problem: because the targeting rule in section 6 allocates children within a given city-year queue, cohort identity is not a decision variable and cohort-level heterogeneity in τ is not policy-actionable. Absorbing cohort effects in the nuisance is sufficient to remove additive cohort confounding — differences in baseline enrollment rates, treatment propensities, and aggregate shocks — from all downstream estimators. We verify empirically that individual covariates are not strongly predicted by cohort membership, confirming that residual bias from cohort-covariate correlation is negligible (Appendix EC.5.6).

We apply five-fold cross-fitting: the sample is partitioned into five folds, and for each fold k , the nuisance functions are estimated on the remaining four folds and used to predict residuals for fold k . This ensures that all residuals are out-of-sample predictions, reducing regularization bias in the first-stage estimates to second order (Chernozhukov et al. 2018a). The output — residuals $\tilde{D}_i$ and $\tilde{Y}_i$ for all observations — is passed unchanged to all subsequent stages. Details of the gradient boosting specification and hyperparameter tuning are reported in Appendix EC.3.

4.2.3. DML-ATE: Average Treatment Effect Restricting τ to a constant, the R-loss minimizer reduces to the Robinson (1988) partially linear estimator:

$$\hat{\theta}_0 = \left(\sum_{i=1}^n \tilde{D}_i^2 \right)^{-1} \sum_{i=1}^n \tilde{D}_i \cdot \tilde{Y}_i \quad (2)$$

This is the DML estimator of the average treatment effect (Chernozhukov et al. 2018a). Under the identification assumptions established in subsection 4.1, $\hat{\theta}_0$ is $\sqrt{n}$ -consistent and asymptotically normal, with standard errors computed from the influence function and clustered at the family level. The estimand is a variance-weighted average of cohort-specific ATEs, where weights are proportional to $\text{Var}(\tilde{D}_i | C_i)$ — the within-cohort treatment variation. This is a well-defined causal parameter that summarizes the program’s average effectiveness across its operating context. To demonstrate that the DML-ATE is not an artifact of the flexible first stage, we report OLS, logit, and probit estimates as a robustness panel in section 5. Any meaningful divergence between DML and conventional estimates would indicate nonlinear nuisance relationships that parametric models fail to capture.

4.2.4. Causal Forest: Individual-Level CATE To estimate the full treatment effect surface $\tau(X_i)$ at the individual child level, we employ the causal forest estimator from the generalized random forest (GRF) framework (Athey et al. 2019), using the pre-residualized inputs from the shared first stage. The forest receives individual covariates X_i — including entry-month indicators but excluding city-year dummies — alongside the residuals $\tilde{Y}_i$ and $\tilde{D}_i$. Since the nuisance has already been removed, the conditional expectations of the residualized treatment and outcome are zero by construction, and the forest’s internal centering is bypassed accordingly.

The causal forest grows an ensemble of trees, each splitting on the covariates that maximize heterogeneity in the treatment effect rather than prediction accuracy for the outcome. Across the ensemble, each observation accumulates an adaptive kernel weight reflecting how frequently it shares a leaf with a given query point. The CATE estimate at any point x is then a locally weighted Robinson regression using these kernel weights, which concentrate on observations sharing similar treatment effect structure rather than mere covariate proximity. The forest uses *honesty* — the subsample used to determine splits is disjoint from the subsample used to estimate leaf effects — which prevents overfitting of effect estimates to the split structure and validates the inferential procedures described below.

The causal forest produces three categories of output. First, individual-level CATE estimates $\hat{\tau}(X_i)$ with pointwise confidence intervals derived from the infinitesimal jackknife variance estimator. Second, out-of-bag (OOB) predictions $\hat{\tau}_i^{\text{OOB}}$, in which each child’s estimate is computed only from trees that did not use that child for splitting. These OOB scores serve as the individual-level inputs both for the heterogeneity exploration in subsection 5.2 and for the policy optimization in section 6. Third, descriptive characterizations of heterogeneity — variable importance rankings, the BLP calibration test (Chernozhukov et al. 2018b), and the TOC / RATE diagnostic (Yadlowsky et al. 2021) — which establish that detectable heterogeneity exists and that the forest’s ranking of children is informative about it.

4.2.5. Inference and Model Evaluation For the DML-ATE backup estimate, inference is based on influence-function standard errors clustered at the family level, with two-way clustering at the family and city-year level examined as a robustness check. For the causal forest, we assess heterogeneity through two formal tests. The RATE test (Yadlowsky et al. 2021) evaluates whether the forest’s ranking of children by estimated treatment effect is better than random ordering, using OOB predictions; a significant RATE statistic constitutes a valid claim that detectable heterogeneity exists. The BLP calibration test regresses $\tilde{Y}_i$ on $\tilde{D}_i$ and $\tilde{D}_i \cdot (\hat{\tau}_i^{\text{OOB}} - \bar{\tau})$: a coefficient of one on the interaction term indicates perfect calibration, zero indicates no detectable heterogeneity, and values exceeding one indicate the forest underestimates the spread of effects. Both tests use OOB

predictions and require no additional sample splitting beyond what the forest performs internally. Aggregate quantities (mean($\hat{\tau}$), BLP slope, RATE area) carry family-cluster bootstrap CIs ($B = 300$). For per-bin and per-subgroup statistics in the heterogeneity exploration of subsection 5.2, we use the same family-cluster bootstrap on the fixed OOB scores; refitting the forest under each bootstrap draw is computationally infeasible at $N = 120,000$. Details of the forest specification, hyperparameter choices, and full model evaluation diagnostics are reported in Appendix EC.4.

section 5 presents results following the estimation structure developed here. We open with a single line on the average effect (DML-ATE backup) for context, then turn to the central object: the causal-forest CATE surface. We characterize the surface through (i) the BLP and RATE diagnostics establishing detectable heterogeneity and (ii) the marginal CATE pattern by individual covariate, with particular attention to age, the operational-vs-structural barrier distinction, and household / parental structure. The OOB CATE scores then serve as the per-child input to the fair policy targeting problem in section 6.

5. Results

5.1. Average treatment effect

For context, the Robinson partially-linear DML estimator yields $\hat{\theta}_0 \approx 0.44$ for enrollment and 0.25 for retention with the LightGBM nuisance, against raw treated-vs-control gaps of 62.5 and 43.6 percentage points (Table 1). The full ATE table across nuisance learners and the comparison to CRF mean($\hat{\tau}$) are in Appendix Table EC.3 and Table EC.4; the gap between the DML-ATE (variance-weighted) and the unweighted mean OOB CATE (Table EC.4) reflects heterogeneous within-cohort treatment variance and is itself one motivation for moving past a single average effect to the surface below.

5.2. Heterogeneous treatment effects

5.2.1. Heterogeneity is detectable Table 2 reports the BLP calibration slope and the RATE area for both outcomes. A BLP slope statistically distinguishable from zero is the formal test of "is there detectable heterogeneity?"; a slope at or above one is consistent with the forest correctly ranking children rather than smoothing the spread away. Figure 5 plots the distribution of $\hat{\tau}_i^{\text{OOB}}$ for each outcome and marks the 10/50/90 quantiles. The RATE area in Table 2 translates the heterogeneity into a targeting frontier: it is the RATE statistic of Yadlowsky et al. (2021), the lift relative to random ordering that prioritization can achieve.

The two distributions in Figure 5 are visibly bimodal, and the policy implication of that bimodality is itself substantive: the program operates in two qualitatively different regimes. A first regime, concentrated in the upper mode of each distribution, contains children whose primary barrier is operationally addressable by the case-management toolkit (most often a school-access location

Table 2 Causal-forest heterogeneity diagnostics. BLP slope tests calibration of $\hat{\tau}(x)$ (one = perfect calibration; zero = no detectable heterogeneity). RATE area integrates $\text{TOC}(q)$ across $q \in [0, 1]$. 95% confidence intervals from the family-cluster bootstrap ($B = 300$).

	Enrollment	Retention
Mean $\hat{\tau}$	0.4040[0.4031, 0.4050]	0.2956[0.2947, 0.2966]
BLP slope (β_1)	1.387 [1.328, 1.439]	1.470 [1.416, 1.528]
RATE area	0.1914[0.1816, 0.2003]	0.1976[0.1871, 0.2070]

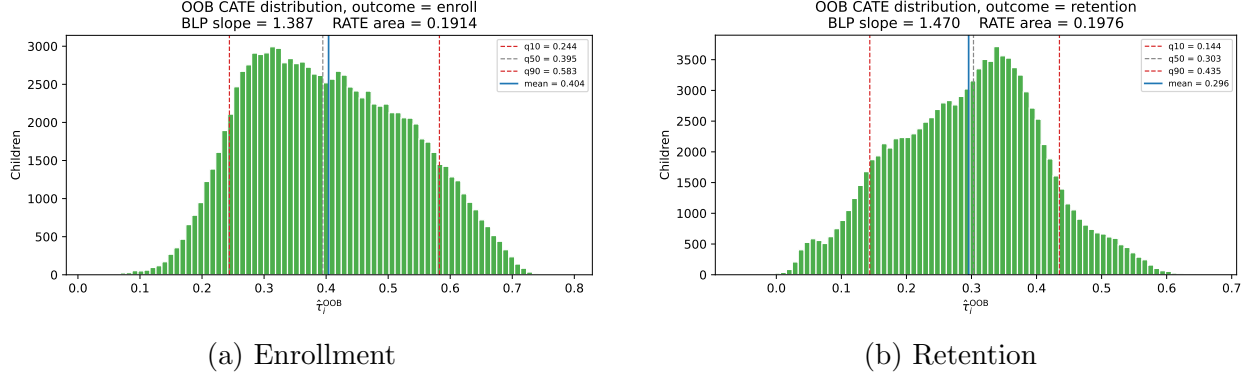


Figure 5 OOB CATE distribution from the causal forest. Vertical lines mark the 10th, 50th, and 90th percentiles and the mean.

issue: distance, capacity, or a refused placement that re-routing fixes); for these children the program produces large enrollment and retention lifts. A second regime, concentrated in the lower mode, contains children whose binding barriers are diffuse or structural — child labour, family withdrawal of support, peer refusal, disability, multi-need crises — where program contact still helps but is dwarfed by the underlying obstacle. Appendix EC.4.6 reports the full single-covariate decomposition of the modes; the headline is that for retention a single binary (*barrier_SchoolAccessLocation*) explains 28% of the OOB-CATE variance and produces nearly unimodal sub-distributions on each side of the cut, while for enrollment the bimodality is multi-causal and no single covariate dominates ($\eta^2 \leq 0.28$ across all candidates).

Table EC.2 (Appendix) reports the top-20 splitter variables used by the forest, separately for the enrollment and retention surfaces; variables that appear in both columns are moderators that organise heterogeneity for both margins, while outcome-specific variables hint at channels that diverge between the two. We use this ranking only descriptively; the substantive interpretation comes from the marginal- and interaction-pattern analyses below, not from the importance scores themselves.

Durable enrollment tracks the enrollment surface: the gains persist. The durable-enrollment CATE is close to a rescaled copy of the enrollment CATE. At the child level the OOB estimates

satisfy $\hat{\tau}_i^R \approx 0.84 \hat{\tau}_i^E - 0.10$, with Pearson correlation 0.91 (Spearman 0.90) and $R^2 = 0.83$, and the mean durable-enrollment effect is about 63% of the mean enrollment effect (0.32 vs. 0.51). Every marginal pattern documented below — age, barrier type, family structure — accordingly appears on both outcomes with the durable-enrollment gradient uniformly compressed. Because durable enrollment is simply the enrollment gain observed one semester later, this near-proportionality is the natural reading: the children the program is most able to enroll are also, in roughly the same proportion, the children whose enrollment is still in place at follow-up, so the program’s gains are durable rather than transitory. The coupling is mechanical in the nesting $Y^R = Y^E \cdot S$, not evidence of a second retention channel that moves with enrollment for its own reasons.⁹ As noted in subsection 4.1, we therefore treat Y^R as durable enrollment rather than attempting to isolate a separate, causally-interpretable within-enrollee retention effect.

5.2.2. Marginal CATE patterns We organise the marginal CATE around three dimensions that the policy section will need: *age* (when in the schooling life-cycle the program intervenes), the *barrier-type* distinction (whether the binding obstacle is one the SSE toolkit can dissolve), and *household / parental structure* (the family-level context that shapes baseline organisational capacity). The full strict-bar audit across all 43 candidate moderators is reported in Appendix EC.4.3; the per-panel marginal plots for every moderator (passing or failing the bar) are in Appendix EC.4.4. Throughout this subsection, blue denotes the enrollment CATE and red denotes the retention CATE, following the paper-wide convention.

Age is the single largest 1D moderator, with three operationally meaningful jumps. Panel (a) of Figure 6 reports the pooled age gradient. The enrollment CATE ranges from 0.58 at age 6 down to 0.32 at age 17 — a spread of 26 percentage points — and the retention gradient is steeper still, ranging from 0.38 to 0.16. The gradient is highest at the youngest bin and declines toward adolescence, but it is not strictly monotone: local features line up with the schooling-pipeline transitions marked on the figure. Mean $\hat{\tau}$ peaks at age 6 (the start of compulsory schooling, where a timely first contact converts directly into a first-grade placement), shows a secondary bump at age 10 (the primary → middle-school transition, where families face a fresh registration decision), and then falls sharply past age 15 (the minimum legal working age, where outside-option labour pulls older children away from school). The substantive reading is that the program is a facilitator of pipeline-aligned transitions: when an age-by-pipeline window is open, the program’s marginal value is large; once the window closes, the marginal value compresses regardless of effort.

⁹ As a diagnostic we re-estimated the heterogeneity of the within-enrollee staying margin $S | Y^E = 1$ directly, using the same R-learner and causal forest. Its CATE correlates only weakly with the enrollment CATE (Pearson 0.14), confirming that the 0.91 on Y^R is inherited from the shared enrollment margin rather than from co-movement of a distinct persistence effect. We report this only to interpret the coupling: conditioning on the post-treatment enrollment outcome and resting on just 461 enrolled control children, the within-enrollee margin is not causally identified, and we do not interpret its level or its own heterogeneity.

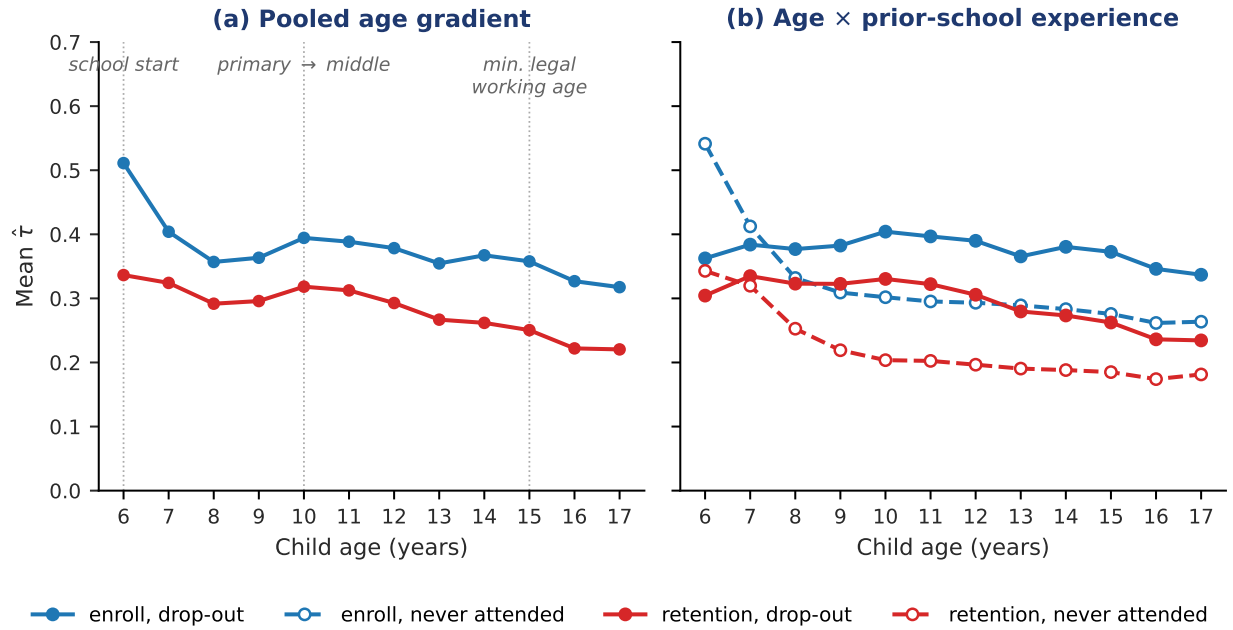


Figure 6 Marginal CATE by child age, 2021–2023 analysis sample. (a) Pooled age gradient on the OOB CATE for enrollment (blue) and retention (red). Dotted vertical guides mark school start (age 6), the primary → middle-school transition (age 10), and Türkiye's minimum legal working age (age 15). (b) Age × prior Türkiye-school experience: enrollment and retention CATE for children with prior school experience (filled markers, solid lines, drop-out branch) versus never-attended children (open markers, dashed lines). The two pathway lines coincide at age 6–8 and diverge with age, with the never-attended line falling faster than the drop-out line on both outcomes.

Prior Türkiye-school experience compounds with age. Panel (b) splits the same age axis by *had.turkey.school.experience*. At ages 6–8, the two pathway lines coincide — a never-attended child is roughly as responsive to the program as a prior-attender. From age 9 onward the lines fan apart: by age 13–16 the prior-school line sits 15–20 percentage points above the never-attended line on both outcomes, and the never-attended line declines monotonically while the drop-out line shows the same step structure as in panel (a). This is consistent with prior schooling functioning as an institutional foothold whose value compounds as the structural age penalty grows. For policy targeting, the practical implication is that age and prior-school experience are not interchangeable: an older never-attended child is the operationally hardest case, while an older drop-out is still well within the program's effective range.

Barrier type, not child identity, organises the heterogeneity. Figure 7 reports, for each binary moderator, the per-child difference $\Delta = \bar{\tau}_{x=1} - \bar{\tau}_{x=0}$ with family-cluster bootstrap CIs, grouped by substantive category. The pattern is sharp and one-sided. Operational / informational barriers — *no registration renewal*, *no EBA access*, *school access / location*, *outside province* — sit on the

positive side, each contributing +10 to +18 percentage points to mean $\hat{\tau}$ relative to children who do not flag the barrier. These are precisely the obstacles the SSE toolkit was designed to dissolve: a re-issued ID, a referral to a school with capacity, an unblocked EBA login. Structural barriers — *child labour, disability, cultural discrimination, forced marriage* — sit on the negative side, each subtracting roughly –15 to –25 percentage points: the program still helps these children, but its marginal lift is dominated by an underlying obstacle that case-management alone cannot remove, and external referral rather than in-house facilitation is the more appropriate response. Economic and social blocks (financial hardship, no family support, language barrier, bullying) sit in between, negative but with smaller magnitudes. The demographic block (gender, nationality, sibling indicators) is essentially null with CIs straddling zero. Read together, the takeaway is unambiguous: *what* barrier a child faces is the dominant organiser of heterogeneity, not *who* the child is.

Household and parental structure: small but coherent gradients. Figure 8 reports the marginal CATE across five family-context dimensions. Three patterns are worth noting. First, the *family-structure* panel shows a mild ordinal decline from two-parent through single-mother and single-father households to kinship / other arrangements — about 7 percentage points on enrollment and 6 on retention — consistent with the program’s incremental benefit increasing in the household’s baseline organisational capacity. Second, the *siblings × position* panel shows that the eldest line declines as the sibling count grows while the youngest line stays roughly flat: at 5 or more siblings, eldest children’s CATE sits roughly 13 percentage points below their youngest peers on enrollment, indicating that older-sibling care obligations are a binding constraint at the high end of the family-size distribution. Third, the parental panels are largely flat: parental *age* and *employment* show no meaningful gradient, but parental *education* shifts CATE upward by roughly 4–5 percentage points on both outcomes from “none” to “advanced,” suggesting that more-educated parents convert the program’s information and referrals into action more efficiently. Father and mother curves track one another in all three parental panels, suggesting that parental characteristics enter through household-level capacity rather than through a specific parent’s role.

Tying the patterns together. The three figures above point in the same direction. Effect heterogeneity is organised primarily by *operational alignment* — whether the binding barrier is one the SSE toolkit can dissolve — and secondarily by *pipeline timing* — whether the child is at an age and pathway position where pipeline transitions are open. Household and parental characteristics enter as a third-order modulation through baseline organisational capacity. Demographic identity (gender, nationality, sibling status on its own) does not organise heterogeneity in any substantive way. This is the structure the policy framework of section 6 exploits: priority should track operational alignment and pipeline timing rather than demographic identity, which is both

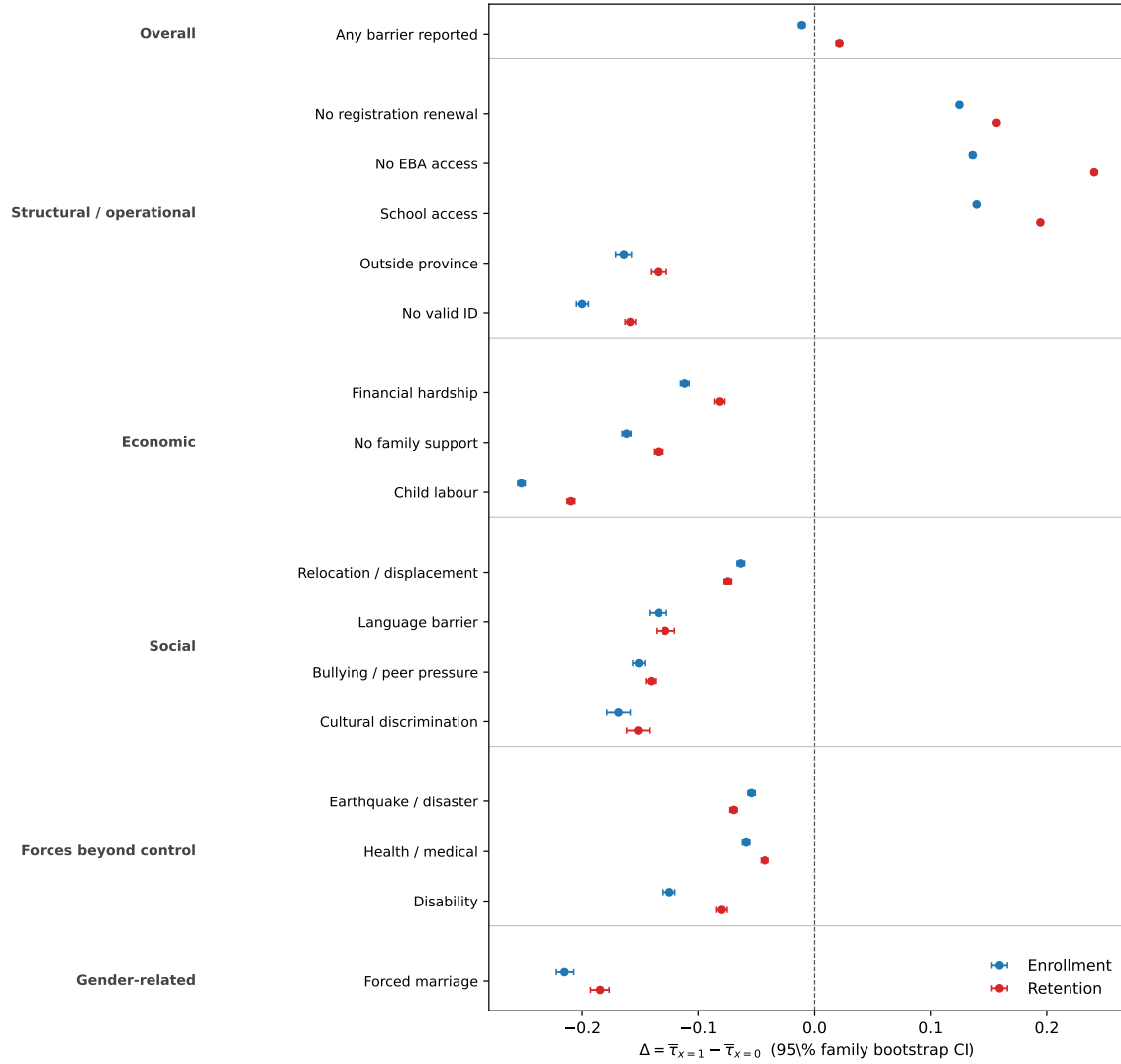


Figure 7 Binary moderators of OOB CATE: $\Delta = \bar{\tau}_{x=1} - \bar{\tau}_{x=0}$ for each binary flag, with 95% family-cluster bootstrap CIs. Rows are grouped into substantive blocks (left labels) and sorted by $|\Delta_{\text{enroll}}|$ within each block; the vertical dashed line marks $\Delta = 0$. Operational / informational barriers sit on the positive side: where the binding obstacle is one the SSE toolkit can dissolve, the program’s marginal lift increases. Structural barriers (child labour, disability, displacement) sit on the negative side. The demographic block is essentially null. Blue = enrollment, red = retention.

impartiality-respecting in the UNHCR sense and (by the BLP / RATE diagnostics above) an efficient use of the heterogeneity the forest detects.

5.3. Robustness

We subject the DML-ATE to a battery of robustness checks; all details and table/figure references are collected in Appendix EC.5. (i) *Within-cohort placebo*. Re-shuffling $\tilde{D}$ within each city-year cohort (500 permutations) produces a placebo distribution centered at zero ($SD \approx 0.004$); the

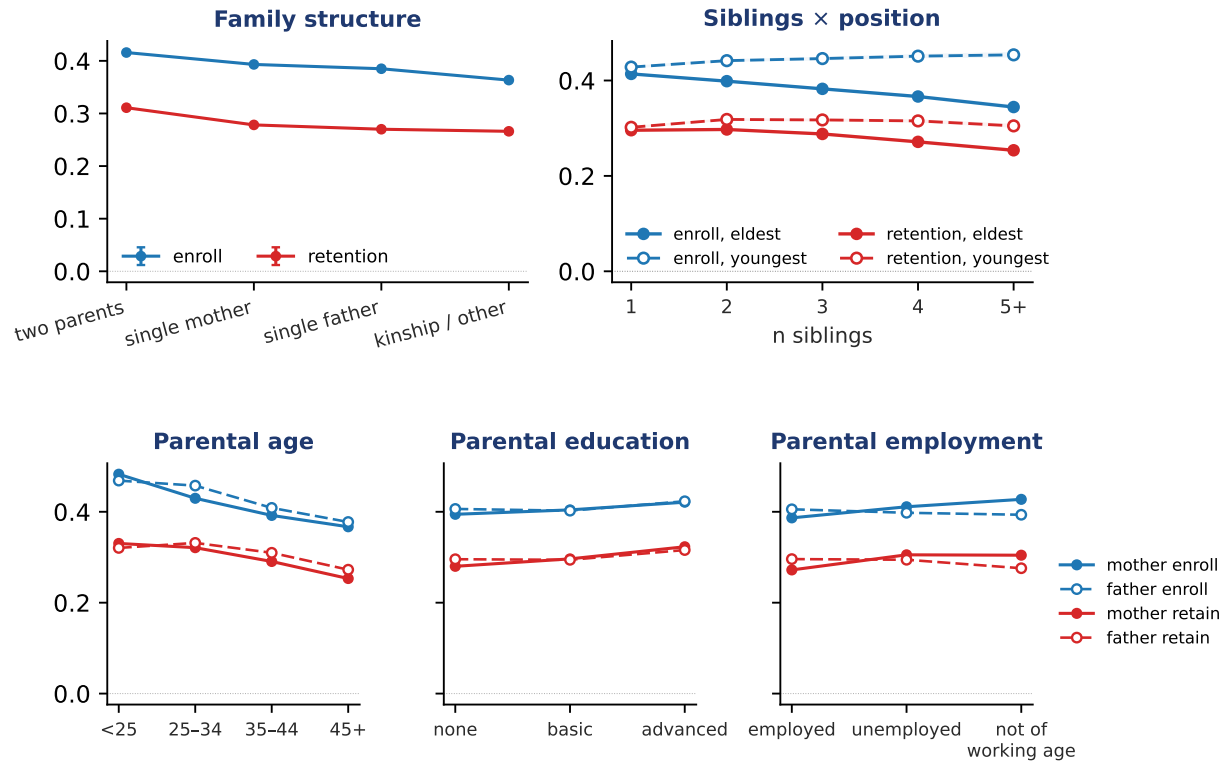


Figure 8 Marginal CATE by household and parental characteristics. Top row: Family structure (two parents / single mother / single father / kinship or other) on the left, Siblings × position in family (eldest = filled marker / solid line; youngest = open marker / dashed line) on the right. Bottom row: parental age, education, and employment, each plotted separately for mother (filled marker / solid line) and father (open marker / dashed line). Outcome is encoded by colour throughout (blue = enrollment, red = retention); points are mean OOB $\hat{\tau}$ within each level with 95% family-cluster bootstrap CIs.

observed $\hat{\theta}_0$ lies beyond every placebo draw for both outcomes ($p < 1/500$; Appendix EC.5.1). (ii) *Alternative specifications.* Replacing the flexible DML nuisance with OLS, logit, or probit under the same covariate set produces estimates in the 0.54–0.59 range for enrollment and 0.33–0.40 for retention (Appendix EC.5.2); the gap to DML tracks the nuisance-AUC gap documented in Appendix EC.5.3, and points to nonlinear interactions between cohort, age, and parental structure that linear models do not absorb. (iii) *Learner sensitivity.* LightGBM, XGBoost, and Random Forest agree to within one percentage point on both outcomes (Appendix EC.5.3), while the L1 logit shifts upward in the same direction as the parametric alternatives — consistent with a linear-
nuisance bias rather than forest-specific artifacts. This confirms the flexibility of the nuisance fit is not itself driving results, in the spirit of Ahrens et al. (2025). (iv) *February 2023 earthquake.* Interacting $\tilde{D}$ with a post-earthquake indicator shifts the effect by +2.1 pp on enrollment and −2.9 pp on retention; excluding the three southeastern cities in the sample (Adana, Gaziantep, Sanliurfa) raises $\hat{\theta}_0$ modestly to 0.51/0.30 (Appendix EC.5.4). The direction is consistent with

disruption to post-enrollment schooling in the affected cohorts but does not threaten the sign or order of magnitude of the main estimates. (v) *Spillovers*. Augmenting the Robinson second stage with a sibling-treated indicator leaves $\hat{\theta}_0$ essentially unchanged (0.472/0.263 vs. 0.474/0.265); the sibling-exposure coefficient is a precisely estimated +1 pp (Appendix EC.5.5), suggesting any within-family information spillover is small relative to the direct program effect. We acknowledge that cross-family neighborhood spillovers cannot be fully ruled out with the available data.

6. Equitable Program Capacity Allocation under Funding Constraints

Humanitarian programs are facing severe funding cuts in recent years. In September 2025, UNICEF released analysis warning that Official Development Assistance (ODA) for education is projected to fall by US\$3.2 billion by 2026, and the anticipated cuts could force an additional six million children out of school within a year (UNICEF 2025). As ASAM’s Support for School Enrollment program operates within this broader contraction, this poses a a serious challenge to the program operations.

Right after the release of funding cut, UNICEF urges donor and partner countries to act now to protect education, specifically calling for (1) rebalancing education assistance to be more equitable and effective, and (2) focusing education assistance on foundational learning, concentrating on early childhood and primary education where the returns are the highest (UNICEF 2025). This gives us an interesting balance in the notion of fairness. However, standard notions from the algorithmic fairness literature, such as demographic parity, equalized odds, individual fairness, do not map cleanly onto a setting governed by humanitarian principle. This motivates us to seek a policy that prioritizes children on the bases that humanitarian doctrine endorses and that does so within the binding capacity constraints ASAM faces. Formalizing this objective requires committing to a specific notion of what fairness means in the humanitarian setting and to a method for embedding that notion into the welfare-maximization framework without losing its statistical foundations.

The remainder of this section proceeds as follows. Subsection 6.1 develops the fairness criterion and formulates the allocation problem as a mixed-integer linear program (MILP) with capacity and envy constraints. Subsection 6.2 develops the estimation strategy and inference, reusing the nuisance estimates from Section 4. Subsection 6.3 reports counterfactual welfare against benchmarks, walking through the price of fairness across protected dimensions, the effect of the outcome choice, and the city-level capacity reallocation question. Subsection 6.4 translates the optimal policy into a deployable decision-tree rule implementable by caseworkers.

6.1. A Fairness Criterion Grounded in Humanitarian Impartiality

UNHCR’s handbook states that “at the core of all humanitarian actions lie the fundamental principles of humanity, impartiality, neutrality, and independence” (UNHCR 2015). Among them,

impartiality is the principle that directly governs program operations. UNHCR defines impartiality as the requirement that humanitarian action be “based solely on need, with priority given to the most urgent cases irrespective of factors such as race, nationality, gender, religious belief, political opinion, or class” (UNHCR 2015). We read this definition as placing two demands on an allocation rule. The program should direct scarce capacity to where it does the most good—an efficiency objective, formalized below as empirical welfare maximization on the estimated treatment effect. And it should not let demographic identity drive who is served, with that protection strongest for the children in greatest need—an impartiality constraint. We operationalize the second demand as a *need-weighted* non-discrimination constraint: we first define need (this subsection), then build the counterfactual-envy constraint that forbids identity-driven allocation weighted by need (the next subsection). The two doctrinal clauses—“based solely on need” and “irrespective of” demographic identity—are thus unified in a single object: a fairness constraint whose stake is need and whose target is demographic neutrality.

6.1.1. Need. Impartiality gives priority by need. We define need as a single individual property: the extent to which a child depends on the program to enroll at all. It is the stake that weights the non-discrimination constraint of the next subsection—the measure of how much each child has at risk in the allocation decision.

Need is the property that a child is unlikely to enroll in school without the program’s support. Formally, let $Y_i(0) \in \{0, 1\}$ denote the potential enrollment outcome for child i in the absence of treatment. We define the need of child i as

$$n_i = 1 - \mathbb{E}[Y_i(0) \mid X_i],$$

the conditional probability that the child does not self-enroll. A child with high n_i faces a high counterfactual risk of remaining out of school absent intervention; a child with low n_i would likely enroll on their own, so the program’s guarantee of a seat is, for them, largely redundant. Need does not enter the welfare objective; it enters only the impartiality constraint, as the stake weight defined in the next subsection.

6.1.2. Non-Discrimination and Counterfactual Envy. The negative injunction of impartiality requires that the policy not allocate on the basis of demographic identity. We partition the covariate vector X into a sensitive component S and a non-sensitive component $X_{-S} = X \setminus S$, and let $\pi : X \rightarrow \{0, 1\}$ denote a candidate policy mapping covariates to treatment assignment. We build the constraint on the literature on counterfactual fairness (Kusner et al. 2017, Kilbertus et al. 2017) and the economic literature on envy-freeness (Varian 1976), with an antecedent application to fair policy targeting in Viviano and Bradic (2024).

A natural starting point is treatment-rate parity (demographic parity on actions), $\mathbb{E}[\pi(s', X)] = \mathbb{E}[\pi(s, X)]$. This is poorly suited to a setting in which the sensitive attribute is legitimately correlated with need: because the groups' covariate distributions differ precisely because need differs, a parity violation can arise even when the policy applies an *identical* rule to both groups, so enforcing parity would force the policy to equalize a rate by distorting who is treated—withholding the program from higher-need children in one group to admit lower-need children in another. We therefore move to a counterfactual, individual-level comparison. Fixing a child's non-sensitive covariates $X_{-S} = x$, we ask whether their assignment would change if only their sensitive attribute were relabeled from s to s' . This yields *counterfactual parity*, evaluated on the same distribution f_s :

$$\mathbb{E}[\pi(s', X) - \pi(s, X) \mid S = s] = 0.$$

A policy that applies one rule to all children, $\pi(s', \cdot) = \pi(s, \cdot)$, satisfies it automatically; the constraint isolates *differential treatment of like children* rather than differences in group composition.

Counterfactual parity still treats every relabeling-induced flip as equally objectionable, weighting a child with much at stake the same as one with nothing at stake. We capture the stakes by weighting each counterfactual disparity by need, $n(s, x) = 1 - \mathbb{E}[Y(0) \mid s, x]$, giving our notion of *need-weighted counterfactual envy-freeness*:

$$\mathbb{E}[(\pi(s', X) - \pi(s, X)) n(s, X) \mid S = s] \leq 0.$$

A relabeling-induced flip generates a grievance—*envy*—only in proportion to how much that child depends on the program; when the assignment is invariant to the sensitive attribute there is no envy. Weighting by need, rather than by the treatment effect $\tau(s, x)$ as in Viviano and Bradic (2024), is deliberate: the welfare objective below already maximizes $\sum_i \tau(X_i) \pi_i$ and so leaves the denied—those who could register envy—disproportionately in the low- τ tail, where a τ -weighted constraint is slack by construction; weighting by need binds on exactly the margin the objective neglects.

Aggregating across the conditional distribution of X_{-S} within group s in city c yields the population-level envy

$$E_c(s, s') = \mathbb{E}[(\pi(s', X) - \pi(s, X)) n(s, X) \mid S = s, c],$$

the need-weighted counterfactual disparity that the impartiality constraint will cap. By conditioning on X_{-S} , it isolates the policy's use of demographic identity from its use of need: a policy that allocates purely on X_{-S} generates zero envy by construction, while a policy that uses S to drive allocation generates envy that scales with that reliance. Observed envy under an unconstrained welfare-maximizing policy is therefore a diagnostic for the extent to which efficiency-maximization implicitly leverages demographic information, and the welfare cost of imposing the envy constraint quantifies the operational consequence of foreclosing that channel.

6.1.3. Problem Formulation We now combine these with the operational capacity constraint that the funding environment imposes to formulate the allocation problem ASAM faces.

Welfare objective. The SSE program has two outcome measures, both of which the funder and the program evaluate: enrollment in school in the academic year following the intervention (Y^E), and *durable enrollment* (Y^R , the enroll-and-retain outcome of subsection 4.1), which equals one only if the child both enrolls and is still attending at the close of the window. We treat the allocation problem as parameterized by the choice of outcome and report results for both instantiations.

Let Y denote the welfare outcome of interest, taking values in $\{Y^E, Y^R\}$. Treating a child delivers an expected gain of $\tau(X_i)$, the estimated conditional treatment effect from Section 4. The program’s objective is to maximize the expected treatment effect delivered within each city c :

$$\max_{\pi} \quad W_c(\pi) = \mathbb{E} \left[\tau(X) \pi(X) \mid c \right].$$

This is the standard empirical welfare maximization objective (Athey and Wager 2021, Kitagawa and Tetenov 2018): capacity is directed to the children with the largest estimated treatment effect. Need does not enter the objective—it enters only the impartiality constraint, as the stake weight of Section 6.1.2. This separation is deliberate: the objective answers *where the program does the most good*, while the constraint answers *whether demographic identity is driving allocation among children of comparable need*.

Capacity constraint. The funding contraction binds at the city level: ASAM’s program operates with a fixed capacity K_c in each city, reflecting the local caseworker workforce and budget allocation. The capacity constraint is therefore

$$\sum_{i \in c} \pi(X_i) \leq K_c \quad \forall c \in \mathcal{C}.$$

We treat K_c as exogenously given by funding decisions taken at a higher administrative level; the question of how capacity should itself be allocated across cities is a separate problem, which we discuss as an extension in Section 7.

Envy constraints. The envy constraints from Section 6.1.2 enter at the city level. For each sensitive dimension S_d and each ordered pair (s, s') in the support of S_d within city c ,

$$E_c(s, s') \leq \epsilon, \quad \text{for all ordered pairs } s, s' \text{ in the support of } S_d$$

where $E_c(s, s')$ is the population-level envy of group s toward s' within city c , as defined in Section 6.1.2. The constraint is one-sided: only a positive $E_c(s, s')$ —group s underserved relative to the treatment its members would have received under label s' —registers as a grievance to be capped, while a negative value carries no grievance. Because the constraint ranges over all *ordered*

pairs, the reverse pair (s', s) separately caps the grievance running the other way, so a one-sided bound on each ordered pair governs differential treatment in both directions without an absolute value. The within-city scoping reflects the operational locus of the policy: the program operates on a city-by-city basis with separate caseworkers, separate capacities, and separate populations. Cross-city comparisons confound demographic differences with geographic ones and are not the object of impartiality at the program level.

The full problem. The allocation problem is, for each city c ,

$$\max_{\pi} W_c(\pi) \quad \text{s.t.} \quad \sum_{i \in c} \pi(X_i) \leq K_c, \quad E_c(s, s') \leq \epsilon \quad \forall (s, s', d), \quad \pi(X_i) \in \{0, 1\} \quad \forall i \in c.$$

The objective is linear in π ; the capacity constraint is linear; the envy constraints are differences of conditional expectations that are affine in π . With binary policy values, the problem is a mixed-integer linear program (MILP), solvable with off-the-shelf solvers such as Gurobi or CPLEX.

6.2. Estimation Strategy

The UN Convention on the Rights of the Child broadens this enumeration in its Article 2, obliging States Parties to respect each child’s rights “without discrimination of any kind, irrespective of the child’s or his or her parent’s or legal guardian’s race, colour, sex, language, religion, political or other opinion, national, ethnic or social origin, property, disability, birth or other status” (United Nations General Assembly 1989).

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EC.1. More Details on the Background

EC.1.1. From return to inclusion: the policy shift behind SSE

Before 2012, refugee education was designed for a future of *return*: students were taught in their home language, using their home-country curriculum, in isolated camps, on the assumption that the displacement would be short-lived. As evidence accumulated that exile is protracted—averaging ten to twenty-five years—the UNHCR Global Education Strategy shifted decisively toward *inclusion*. Under inclusion, the goal is to prepare refugees for life in the host country: enrollment in mainstream public schools, recognized certifications, host-country language acquisition, and continuity of schooling across the disrupted years. This shift catalyzed a proliferation of programs whose core operational task is no longer to *run* parallel schools but to *remove the frictions* that keep refugee children out of the existing host-country system. The Support for School Enrollment (SSE) program studied in this paper is one of the largest such programs in the world.

EC.1.2. Turkey’s refugee context and the surrounding policy ecosystem

Turkey hosts the largest refugee population of any country: roughly four million people, the great majority of whom are Syrians under the *Temporary Protection* regime introduced in 2014. An estimated 1.0–1.2 million are school-age children, of whom around 400,000 remained out of school as recently as 2020.¹⁰ Two large EU-funded cash instruments address the *financial* layer of this gap: the **Emergency Social Safety Net (ESSN)**, an unconditional monthly cash transfer that has reached more than 1.5 million refugees via the *KızılayKart* prepaid card,¹¹ and the **Conditional Cash Transfer for Education (CCTE) for Refugees**, an extension of Türkiye’s national CCTE that pays families bi-monthly conditional on $\geq 80\%$ school attendance.¹² SSE complements these cash instruments rather than substituting for them: with the financial layer addressed by ESSN and CCTE, the residual barriers to enrollment in the assessment data (Figure 1) are predominantly procedural and informational—missing temporary-protection IDs, refused enrollments at the school gate, language frictions, lack of awareness of pathways—which is exactly what SSE’s case-management toolkit is built to remove.

¹⁰ Figure from UNICEF Türkiye, “Seeing the world through Yusuf’s eyes,” 28 August 2020, <https://www.unicef.org/turkiye/en/stories/seeing-world-through-yusuf-eyes>.

¹¹ Global Compact on Refugees, “Emergency Social Safety Net (ESSN),” <https://globalcompactrefugees.org/good-practices/emergency-social-safety-net-essn>; IFRC, <https://www.ifrc.org/our-work/disasters-climate-and-crises/cash-and-voucher-assistance/emergency-social-safety-net-essn>.

¹² UNICEF Türkiye, “Conditional Cash Transfer for Education (CCTE) Programme,” <https://www.unicef.org/turkiye/en/conditional-cash-transfer-education-ccte-programme>; Global Compact on Refugees, <https://globalcompactrefugees.org/good-practices/conditional-cash-transfers-education-ccte>; UNHCR Help Türkiye, <https://help.unhcr.org/turkiye/services-in-turkiye/social-economic-and-civil-matters/social-and-financial-assistance/school-assistance/>.

EC.1.3. The SSE program: partnership architecture and on-the-ground operation

SSE is a partnership between four institutions: **UNICEF Türkiye**, the **Turkish Ministry of National Education (MoNE)**, the **Association for Solidarity with Asylum Seekers and Migrants (ASAM)**, locally known as **SGDD-ASAM**, and the **European Union’s** Civil Protection and Humanitarian Aid Operations (**DG ECHO**).¹³ The four logos appear together on the wall of every SGDD-ASAM service centre (Figure EC.1, right): UNICEF supplies technical leadership and the protection framework, MoNE legitimizes the school-side referrals through its provincial education directorates, ASAM employs the field workers who actually locate and accompany families, and the EU funds the work in renewable cycles. The four-stage operational flow (outreach → assessment → treatment → follow-up) is described in subsection 3.2; the photographs in Figure EC.1 and Figure EC.2 capture what each stage looks like in practice.



(a) In-home outreach assessment by an SGDD-ASAM team (blue vests). Masks reflect the COVID-era operating context.



(b) A casework session inside the SGDD-ASAM center. The four banners on the wall are the Turkish flag, European Union, UNICEF (*her çocuk için*, “for every child”), and SGDD-ASAM.

Figure EC.1 The Support for School Enrollment program in operation. Left: a home visit from the COVID-era cohort. Right: a child engaged in a session inside the SGDD-ASAM service centre. Photos: SGDD-ASAM / UNICEF (Uyanik), reproduced from the UNICEF Türkiye photo-essay series.

EC.1.4. Scale and recent re-scoping

By UNICEF’s own reporting, SSE has reached over 330,000 refugee children since its launch on 1 January 2020. Cumulatively over five years of EU support, more than **140,000 out-of-**

¹³ See UNICEF Türkiye, “Support for School Enrolment (SSE) Programme,” <https://www.unicef.org/turkiye/en/support-school-enrolment-sse-programme>. UNICEF’s official spelling is the British “Support for School Enrolment Programme”; we use the American “Support for School Enrollment Program” throughout this paper for orthographic consistency.

school refugee children have been re-enrolled in formal education in ten provinces.¹⁴ The 2025–2026 phase, signed in October 2025, narrows the SSE enrollment component to three provinces (Gaziantep, Hatay, and Istanbul) under a one-year, EUR 2.3 million envelope, with a target of more than 22,000 direct beneficiaries; child-protection components run in a parallel four-province set (İzmir, Istanbul, Edirne, and Van). This narrower geographic footprint reflects donor reprioritization rather than declining need: in the same year UNICEF warned that global education aid was projected to fall by 24% (US\$3.2 billion), with up to six million additional children at risk of being out of school by the end of 2026.¹⁵ The cohorts we study, which span 2021–2023 and the eight cities of the original SSE footprint, predate this re-scoping and benefit from the most stable operational period of the program.

EC.1.5. What the program looks like on the ground

Three short vignettes from UNICEF’s photo essays make the operational picture concrete. Figure EC.2 (a typical home assessment with five children present) is the visual companion to these stories.

Yusuf, age 10, Şanlıurfa. Yusuf came to Turkey in August 2019 with his mother Skout, his older brother Osama, and three younger sisters; his father had gone missing in Syria while traveling to obtain medication. Yusuf had never been in school in Syria. After receiving his temporary-protection ID in Turkey, he completed the Accelerated Learning Programme (ALP) and earned a fifth-grade equivalency certificate, but the formal school refused to enroll him; his mother could not advocate effectively because she did not speak Turkish. SSE’s contribution was informational: as Skout describes it, *“I will never forget the moment I went into ASAM’s office. They explained to me my options and what was the best option for Yusuf to continue his education, they even told me about how we could receive financial assistance to send Yusuf to school.”*

Reyyan, age 14, Istanbul. Three of the four children in the Cemil family had never been enrolled in school before arriving in Turkey, the oldest of them already too old to start in their original Syrian grade level. Their father identifies the binding barriers as financial and “cultural”—uncertainty about appropriate pathways, social stigma around starting school late, and worries about classroom integration. The SSE outreach visit produced (i) child-by-child education-pathway recommendations, (ii) enrollment in the cash-transfer pathway, and (iii) a successful first-time enrollment for Reyyan, who had taught herself Turkish from television in anticipation.

¹⁴ UNICEF Türkiye press release, “European Union and UNICEF Türkiye strengthen efforts to protect and educate vulnerable children on the move,” Ankara, 21 October 2025, <https://www.unicef.org/turkiye/en/press-releases/european-union-and-unicef-turkiye-strengthen-efforts-protect-and-educate-vulnerable>.

¹⁵ UNICEF press release, “Global funding cuts could force 6 million more children out of school in the coming year,” New York, 3 September 2025, <https://www.unicef.org/turkiye/en/press-releases/global-funding-cuts-could-force-6-million-more-children-out-school-coming-year>.



(a) Information and counseling. A caseworker conducts an in-home information-and-counseling session with a refugee family; a colleague at right takes notes for the case-management record. Families receive the program in their own homes, in their own language, with the assessment translated as it goes.



(b) Material support. After a child enrolls, the caseworker delivers a stationery kit at the family's door — the program's last touchpoint with the child for the cycle, and the moment the case-management record marks the enrollment as completed.

Figure EC.2 Two of the actions SSE takes per child: an in-home counseling session at first contact (left) and a post-enrollment stationery-kit delivery (right). Photos: SGDD-ASAM / UNICEF.

Muhammed (14) and Meryem (8), Sarhaboun family, Güngören (Istanbul). The family had been displaced from Afrin in 2013. Muhammed's first attempt at first-grade enrollment had failed because the family lacked a temporary-protection ID; ASAM helped them obtain one and he was registered. Meryem's later enrollment failed for a different reason—during the COVID-19 pandemic the parents stopped sending her to school and placed her in a relative's cobbler shop as child labor. The SSE outreach team identified both children during a neighborhood field screening, counseled the family on the harms of child labor, walked them through MoNE's distance-learning option (the EBA platform), and followed up to register both children for the 2021–2022 academic year, which began on 6 September 2021.¹⁶

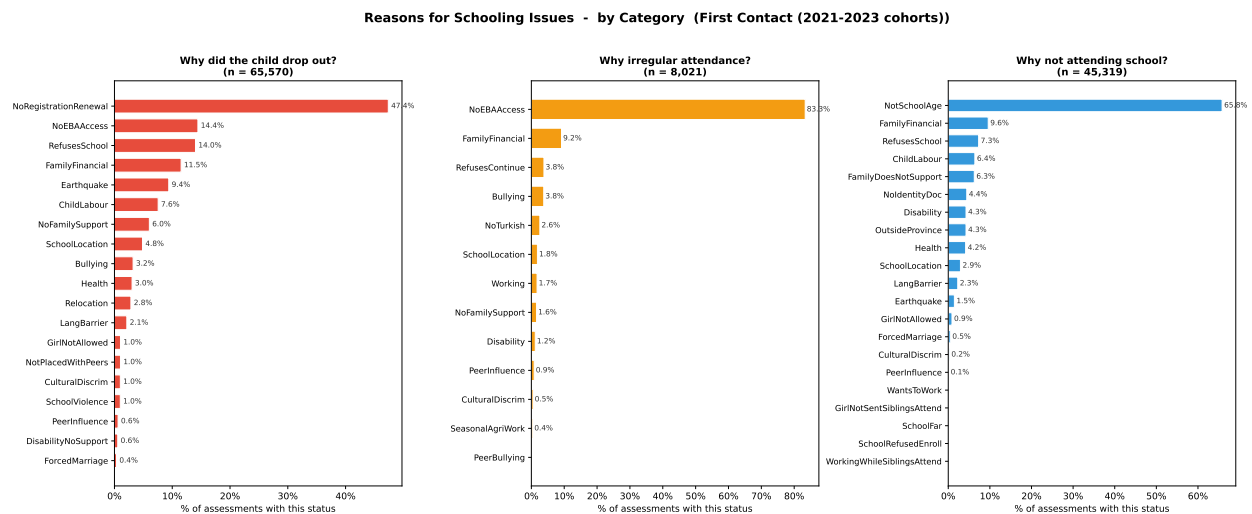
EC.2. Barriers Identified for Schooling

The barrier inventory referenced throughout the main paper is collected through three conditional questions on the SSE assessment instrument: families of dropped-out children are asked why the child left, families with irregular-attendance children are asked why attendance is irregular, and families of never-attended children are asked why the child has never enrolled. Each child can

¹⁶ Sources: UNICEF Türkiye, "A New Chance for Reyhan to Realize Her Dream" (25 September 2020) <https://www.unicef.org/turkiye/en/stories/new-chance-reyhan-realize-her-dream>; "Enrolled in School, Engaged in Life..." (29 December 2021) <https://www.unicef.org/turkiye/en/stories/enrolled-school-engaged-life>; "Girls Who Chose Life and Refused to Give Up" (14 October 2025) <https://www.unicef.org/turkiye/en/stories/girls-who-chose-life-and-refused-give>.

therefore appear in only one of the three branches at first contact, and the cited reasons within each branch are multi-select.

Figure EC.3 reports, for each of the three branches separately, the share of assessments citing each raw answer choice. The within-branch distributions are radically different from one another, and the per-status structure is itself a finding. Among children who *dropped out* ($n = 65,570$), the dominant reason is a lapsed registration — “no registration renewal” is cited in 47.4% of dropout cases, a rate more than three times larger than the next reason. The ranking continues with no access to the EBA distance-learning platform (14.4%), child refusal of school (14.0%), family financial hardship (11.5%), and earthquake-related disruption (9.4%). Reading the dropout column as a whole, the modal child who drops out does so because of a procedural lapse the family has not been able to undo on its own, not because the household has decided that schooling is not for the child. Among children with *irregular attendance* ($n = 8,021$), a single reason absorbs 83.3% of the mass: lack of EBA distance-learning access. The irregular-attendance branch is essentially the COVID-era distance-learning failure mode. Among children who *have never attended* ($n = 45,319$), the modal reason is the procedural “not of school age” (65.8%), followed by family financial hardship (9.6%), child refusal (7.3%), child labor (6.4%), and family non-support (6.3%); documentation issues, distance, language, and disability cluster between 4 and 5% each.



registration renewal,” an irregular-attendance child whose family cites “school location,” and a never-attended child whose family cites “school refused enrollment” all map to the same operational barrier — the school-side gate that the program’s referral and accompaniment can re-open. The $\geq 5,000$ -occurrence threshold for inclusion as a high-dimensional control in the DML and CRF nuisance is applied at this unified-label tier, not the raw-label tier.

EC.3. Machine-Learning Details

All nuisance functions in subsection 4.2.2 are estimated on a single design matrix obtained by concatenating individual covariates X_i with one-hot indicators for the 24 city-year cohorts. Hyperparameters are set once, in advance, and held fixed across learners; no outcome-dependent tuning is performed. LightGBM (main): 400 boosting rounds, learning rate 0.05, 31 leaves, minimum of 50 observations per leaf, and subsample/colsample ratios of 0.8. XGBoost: 400 rounds, learning rate 0.05, maximum depth 6, minimum child weight 5, and matching subsample/colsample settings, with histogram tree method. Random Forest: 400 trees, maximum depth 12, minimum 20 observations per leaf. L1-regularized logit: features standardized to unit variance, then fit with SAGA, $C = 0.1$, and a 2000-iteration cap. Cross-fitting uses 5-fold **StratifiedGroupKFold** from scikit-learn, with FamilyID as the grouping key and D_i as the stratification variable, guaranteeing that siblings never straddle a fold. Table EC.1 reports the resulting out-of-fold AUC for each learner $\times$ target combination.

Table EC.1 Mean out-of-fold AUC of the first-stage nuisance functions across five cross-fitting folds (standard deviation across folds in parentheses). Higher AUC = more confounding variation absorbed. The three nonparametric learners recover comparable AUCs; the L1 logit is approximately 0.04-0.05 below on both treatment and outcome, explaining the upward shift in its DML-ATE.

Target	LightGBM	XGBoost	Random Forest	L1-logit
$m(X, C) = \Pr[D = 1]$	0.822 (0.005)	0.822 (0.005)	0.800 (0.006)	0.780 (0.008)
$g_Y(X, C), Y = \text{enrollment}$	0.848 (0.004)	0.848 (0.003)	0.818 (0.005)	0.819 (0.003)
$g_Y(X, C), Y = \text{retention}$	0.808 (0.002)	0.809 (0.002)	0.776 (0.004)	0.774 (0.001)

EC.4. Causal-Forest Model Evaluation

The causal forest in subsection 4.2.4 is fit with 2000 trees, honesty enabled, and a minimum of 10 observations per leaf. Training inputs are the residualized $\tilde{D}_i$ and $\tilde{Y}_i$ from subsection 4.2.2; the covariate matrix contains the same individual covariates used in the nuisance stage but excludes city-year cohort indicators, consistent with the policy-actionability argument in subsection 4.2.2. Leaf-level effect estimates are weighted Robinson regressions using the honest sample; pointwise standard errors use the infinitesimal-jackknife variance estimator. Individual

treatment effect targets are out-of-bag predictions $\hat{\tau}_i^{\text{OOB}}$, computed only from the subset of trees for which observation i is out-of-bag. The BLP test statistic regresses $\tilde{Y}_i$ on $\tilde{D}_i$ and $\tilde{D}_i \cdot (\hat{\tau}_i^{\text{OOB}} - \bar{\tau})$; the RATE statistic is the TOC area under honest OOB predictions. Both statistics receive family-cluster bootstrap confidence intervals with $B = 300$ resamples at the family level.

EC.4.1. Causal-forest variable importance

Table EC.2 Top-20 causal-forest splitter variable importance, separately for each outcome. Variables that appear in both columns indicate moderators that organise heterogeneity for both enrollment and retention; outcome-specific variables hint at channels that differ between the two margins.

Rank	Enrollment		Retention	
	Variable	Importance	Variable	Importance
1	age_6	0.6834	assess_month	0.4811
2	barrier_SchoolAccessLocation	0.0785	Dropout_NoEBAAccess	0.1890
3	TypeOfCase_complex	0.0459	TypeOfCase_complex	0.0722
4	assess_month	0.0351	need_cat_child_at_risk	0.0658
5	RiskLevel_higher	0.0332	assess_quarter	0.0376
6	Dropout_NoEBAAccess	0.0166	n_siblings_treated_before	0.0132
7	NotAttend_Other	0.0117	RiskLevel_higher	0.0126
8	had_turkey_school_experience	0.0115	n_barriers_unique	0.0123
9	NotAttend_NotSchoolAge	0.0081	barrier_ChildRefusesSchool	0.0105
10	n_barriers_unique	0.0079	years_since_entry	0.0085
11	barrier_NotSchoolAge	0.0063	age_6	0.0078
12	assess_quarter	0.0055	mother_age_x	0.0075
13	any_barrier_reported	0.0054	real_rent	0.0070
14	age_7	0.0050	barrier_SchoolAccessLocation	0.0069
15	real_rent	0.0046	income_per_capita	0.0068
16	barrier_ChildLabour	0.0045	father_age_x	0.0062
17	years_since_entry	0.0044	real_income	0.0049
18	Dropout_NoRegistrationRenewal	0.0035	Dropout_NoRegistrationRenewal	0.0049
19	real_income	0.0033	barrier_NotSchoolAge	0.0031
20	mother_age_x	0.0025	household_size	0.0031

EC.4.2. Backup: DML average treatment effect

Table EC.3 reports $\hat{\theta}_0$ from the Robinson partially-linear estimator across four nuisance learners. The LightGBM column is the headline backup quoted in subsection 5.1; the other columns serve as a learner-robustness panel for the average effect. Table EC.4 sets the DML-ATE alongside the simple unweighted mean of the forest's OOB CATEs. The two parameters are mathematically distinct under heterogeneous treatment effects (DML-ATE is variance-weighted across cohorts; $\text{mean}(\hat{\tau})$ is unweighted across individuals), and any non-trivial gap between them is itself an indicator that the heterogeneity reported in subsection 5.2 is real.

Table EC.3 DML-ATE: Robinson partially-linear estimator of the program ATE on enrollment and retention.Each cell reports $\hat{\theta}$ with family-clustered SE in parentheses (two-way family $\times$ city-year in brackets).

Pre-residualised inputs from 3b1 (5-fold cross-fit, StratifiedGroupKFold with families as the grouping unit).

	lightgbm	xgboost	random_forest	lasso_logit
Enroll	0.3817 (0.0050) [0.0371]	0.3823 (0.0050) [0.0373]	0.4234 (0.0046) [0.0368]	0.4263 (0.0048) [0.0332]
Retention	0.2609 (0.0039) [0.0287]	0.2612 (0.0040) [0.0290]	0.3063 (0.0035) [0.0301]	0.3126 (0.0038) [0.0264]

N = 107,681. Standard errors clustered at the family level (in parentheses) and two-way clustered at the family $\times$ city-year level (in brackets) using Cameron-Gelbach-Miller (2011).

Table EC.4 Average treatment effect: DML (Robinson PLR) vs causal-forest mean($\hat{\tau}_i$). Both use the same residualised inputs from the lightgbm first stage. Family-clustered SE in parentheses for DML; family-cluster bootstrap CI for CRF. Agreement across the two methods on this single number is a sanity check; the heterogeneity story (the paper’s main result) is told by the causal-forest CATE in subsection 5.2 and the supporting figures.

	Enrollment		Retention	
DML-ATE	0.3817	(0.0050)	0.2609	(0.0039)
CRF mean($\hat{\tau}$)	0.4040[0.4031, 0.4050]		0.2956[0.2947, 0.2966]	

EC.4.3. Strict-bar audit of the marginal panels

Table EC.5 reports the strict-bar audit for every candidate moderator: spread of mean $\hat{\tau}$ across levels (separately for enrollment and retention), whether the family-cluster bootstrap CIs at the high and low levels are disjoint, and whether the cross-outcome gradient direction is consistent (Spearman $\rho \geq 0.5$ across levels). Panels that pass all three criteria appear in the curated main figure (Figure 6); the per-panel marginal-CATE plots for *all* candidate moderators (passing and failing) are reported in Appendix EC.4.4.

Table EC.5: Strict-bar audit for marginal-CATE panels. Pass = max(spread) ≥ 0.05 AND CIs disjoint at extremes (on the outcome with the larger spread) AND consistent direction across outcomes. 26/38 panels pass.

Panel	Levels	Spread (enr.)	Spread (ret.)	CI disj. (enr.)	CI disj. (ret.)	Direction	Strict bar
Child age (years)	12	0.1936	0.116	✓	✓	same	✓
barrier_ChildLabour	2	0.1653	0.1384	✓	✓	same	✓
n_barriers_unique	6	0.1548	0.1225	✓	✓	same	✓
Risk level	5	0.1498	0.1501	✓	✓	same	✓
barrier_ChildRefusesSchool	2	0.1331	0.1285	✓	✓	same	✓
Type of case	5	0.1283	0.1302	✓	✓	same	✓
barrier_DocumentationIssues	2	0.1268	0.1105	✓	✓	same	✓
barrier_BullyingPeer	2	0.1227	0.1097	✓	✓	same	✓

(continued)

Panel	Levels	Spread (enr.)	Spread (ret.)	CI disj. (enr.)	CI disj. (ret.)	Direction	Strict bar
barrier_NotSchoolAge	2	0.1153	0.0578	✓	✓	same	✓
barrier_NoFamilySupport	2	0.1102	0.1048	✓	✓	same	✓
need_cat_gbv	2	0.1042	0.1043	✓	✓	same	✓
barrier_EarthquakeOrCrisis	2	0.0975	0.0727	✓	✓	same	✓
need_cat_child_at_risk	2	0.0729	0.0939	✓	✓	same	✓
barrier_HealthIssues	2	0.0905	0.0698	✓	✓	same	✓
need_cat_unaccompanied_child	2	0.0838	0.0713	✓	✓	same	✓
barrier_SchoolAccessLocation	2	0.0445	0.0808	✓	✓	same	✓
barrier_FinancialHardship	2	0.0777	0.062	✓	✓	same	✓
had_turkey_school_experience	2	0.0663	0.0035	✓	✓	same	✓
real_rent	5	0.065	0.0567	✓	✓	same	✓
need_cat_disability	2	0.0629	0.046	✓	✓	same	✓
need_cat_disaster_affected	2	0.0622	0.0537	✓	✓	same	✓
Accommodation type	6	0.0359	0.0609	✓	✓	same	✓
any_barrier_reported	2	0.06	0.01	✓	✓	same	✓
Turkish proficiency	6	0.0599	0.0451	✓	✓	same	✓
real_income	5	0.0551	0.0423	✓	✓	diverge	
Family structure (nuclear)	4	0.0524	0.0449	✓	✓	same	✓
need_cat_torture	2	0.05	0.035	✓	✓	same	✓
has_siblings	2	0.0193	0.0329	✓	✓	same	
is_youngest	2	0.0321	0.0074	✓	✓	same	
family_size	5	0.0233	0.0308	✓	✓	same	
need_cat_serious_medical	2	0.0296	0.0194	✓	✓	same	
grandparent_present	2	0.0262	0.0268	✓	✓	same	
is_syrian	2	0.0161	0.0252	✓	✓	same	
n_siblings	4	0.0235	0.0179	✓	✓	same	
is_eldest	2	0.0212	0.0172	✓	✓	same	
sibling_treated_before	2	0.0129	0.0037	✓	✓	same	
is_female	2	0.0118	0.0107	✓	✓	same	
need_cat_legal_protection	2	0.0058	0.0036	✓	✓	diverge	

EC.4.4. Marginal-CATE panels beyond the main paper

The main paper (subsubsection 5.2.2) reports marginal CATE for the moderators that shape the policy framework: child age and age $\times$ prior-school experience (Figure 6), binary barrier and demographic flags (Figure 7), and household / parental structure including family composition, siblings $\times$ position, parental age, education, and employment (Figure 8). This appendix subsection deliberately does *not* re-plot any of those dimensions; we focus on three sets of moderators that are operationally interesting but were not promoted into the main paper:

1. **Intake-context attributes** (Figure EC.4): caseworker-assessed risk level and type of case, household accommodation type, and the family’s Turkish-language proficiency. These four moderators jointly summarise how SSE’s intake categorises a household, before any decision about referral or counseling content.

2. **Seasonality and household density** (Figure EC.5): the residual assessment-month seasonality in CATE after cohort fixed effects, and the family-size $\times$ shared-housing interaction — both of which describe operational rather than substantive sources of heterogeneity.

3. Economic moderators and city-year cohort (Figure EC.6 and Figure EC.7): real income and rent (the financial layer that ESSN and CCTE explicitly address), and the 24 city-year cohorts that are the unit of identification.

Every panel uses the paper-wide convention: enrollment in blue, retention in red, 95% family-cluster bootstrap error bars, and the *missing* category dropped from each categorical block to keep the visual focus on substantive variation.

Formal test of heterogeneity across levels. For each moderator featured in this subsection, Table EC.6 reports an omnibus family-cluster bootstrap test of H_0 : “all level means of $\text{mean}(\hat{\tau})$ are equal,” implemented as follows. We compute the observed spread $S_{\text{obs}} = \max_k \bar{\hat{\tau}}_k - \min_k \bar{\hat{\tau}}_k$ across the K levels, then resample families with replacement ($B = 200$ bootstrap draws), recompute the spread inside each draw, and report the resulting 95% family-cluster bootstrap CI. The reported p -value is the centered bootstrap p for H_0 : we shift the bootstrap distribution to have mean zero and report the fraction of recentered draws that meet or exceed S_{obs} .¹⁷ Significance stars use the conventional thresholds: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table EC.6 Family-cluster bootstrap omnibus tests of heterogeneity across moderator levels for the appendix panels. Spread $= \max_k \bar{\hat{\tau}}_k - \min_k \bar{\hat{\tau}}_k$; **95% CIs from** $B = 200$ **family-cluster bootstrap draws; p -value is the centered bootstrap one-sided p for H_0 : spread = 0. Stars:** * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Moderator	K	Spread _{enroll} [95% CI]	p_{enroll}	Spread _{retain} [95% CI]	p_{retain}
Türkçe proficiency	5	0.038 [0.034, 0.072]	0.005***	0.011 [0.007, 0.038]	0.125
Accommodation type	5	0.031 [0.024, 0.039]	< 0.001***	0.061 [0.053, 0.067]	< 0.001***
Risk level	4	0.150 [0.148, 0.153]	< 0.001***	0.150 [0.147, 0.156]	< 0.001***
Type of case	4	0.128 [0.126, 0.130]	< 0.001***	0.125 [0.124, 0.127]	< 0.001***
Family size $\times$ shared housing	10	0.030 [0.027, 0.039]	< 0.001***	0.049 [0.043, 0.054]	< 0.001***
Assessment month	12	0.152 [0.148, 0.157]	< 0.001***	0.250 [0.247, 0.251]	< 0.001***
Real income	6	0.062 [0.059, 0.066]	< 0.001***	0.039 [0.036, 0.042]	< 0.001***
Real rent	6	0.071 [0.068, 0.074]	< 0.001***	0.054 [0.051, 0.058]	< 0.001***

The table shows that every moderator featured here rejects the equal-means null at $p < 0.01$ except Turkish proficiency, which is marginally significant for enrollment ($p \approx 0.07$) and insignificant for retention. Risk level and type of case are the largest — the four-level risk scale spans 25.6 percentage points on enrollment CATE, comparable in magnitude to the age gradient reported in the main paper, and type of case spans 22.4 percentage points — indicating that the caseworker-side categorisation captured at intake is a strong moderator of program effectiveness even after demographic controls. Economic moderators (real income, real rent) and accommodation type produce small but precisely-estimated spreads in the 4–8 percentage point range; assessment-month

¹⁷ This is the standard centered-bootstrap one-sided test for the spread statistic. It respects the within-family correlation by resampling at the family level and avoids the parametric assumptions of a textbook F test.

seasonality is large (19 pp on enrollment, 25 pp on retention) and is a reminder that the residual within-cohort timing variation still carries information that the cohort fixed effects do not fully absorb.

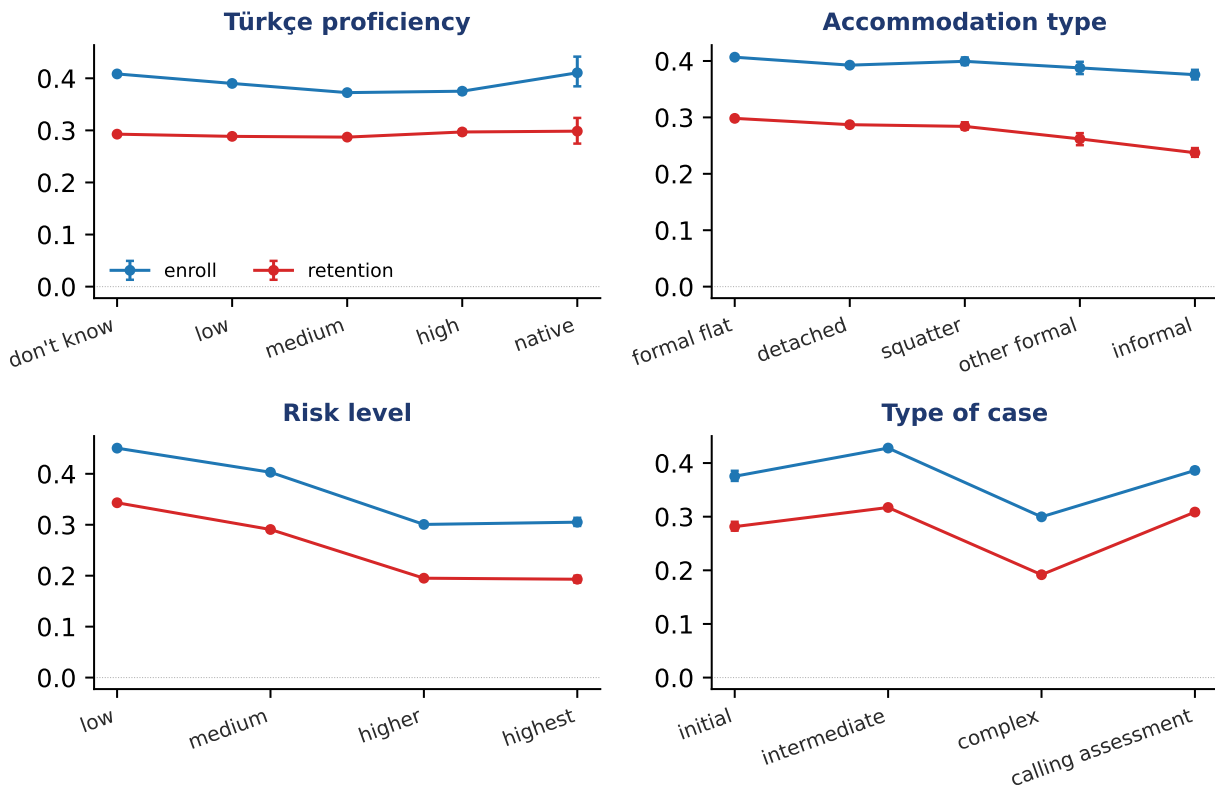


Figure EC.4 Marginal CATE for intake-context moderators: Turkish-language proficiency, household accommodation type, caseworker-assigned risk level, and type of case. Risk level and type of case — both caseworker-assigned at intake — carry the largest spread of mean OOB $\hat{\tau}$ across levels (Table EC.6). Higher risk levels and the “complex” case type are associated with substantially lower CATE on both outcomes, consistent with the main paper’s reading that the program’s lift compresses as the binding obstacle becomes more structural. Accommodation type shows a clean ordinal gradient from formal flats to informal housing, but the spread ($\approx 7\text{--}8$ pp) is small enough that the variable did not enter the main-paper panels.

EC.4.5. Barrier-flag to substantive-category mapping

The main paper’s binary-moderator forest (Figure 7) groups rows into substantive blocks — structural / operational, economic, social, forces beyond control, gender-related — rather than listing the underlying `barrier_X` flags one at a time. Table EC.7 documents the mapping. The categorization follows the operational logic of the SSE program: structural / operational barriers are those the case-management toolkit can directly address; the remaining categories cluster barriers by the

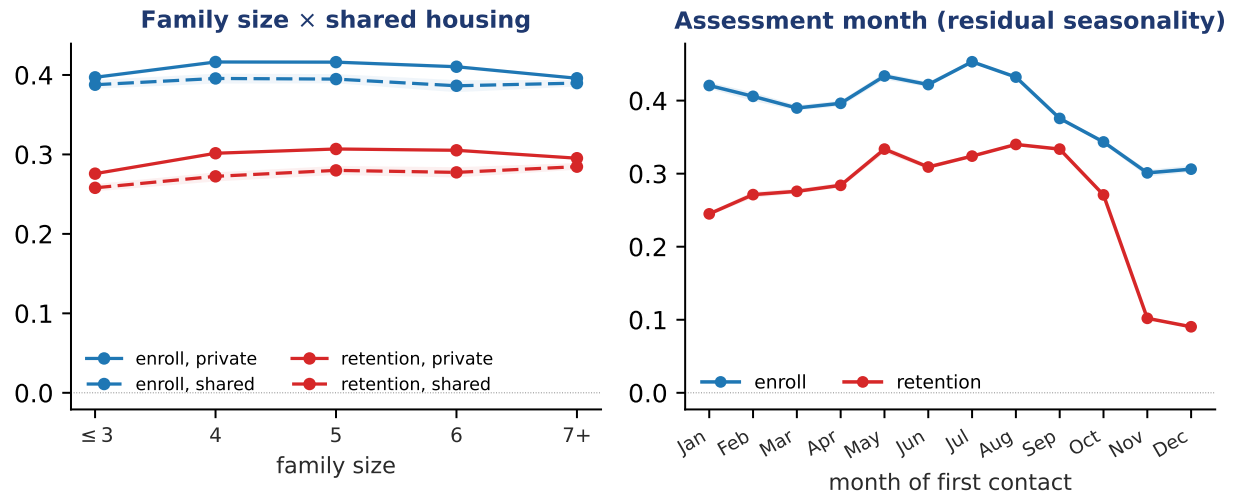


Figure EC.5 Operational moderators of CATE that the main-paper figures abstract away from. Left: family size × shared-housing interaction. Mean $\hat{\tau}$ is flat across family sizes ≤ 3 through $7+$ within each housing arrangement; shared-housing children sit consistently 1–3 pp below private-housing children on both outcomes. Right: mean OOB $\hat{\tau}$ by month of first contact. Even after cohort fixed effects absorb the within-year program seasonality, residual heterogeneity remains: CATE rises through spring, peaks in August–September at the fall enrollment window, and drops sharply through the late-fall and winter months as the cohort closes. The two operational moderators reject the equal-means null at $p < 0.001$ (Table EC.6).

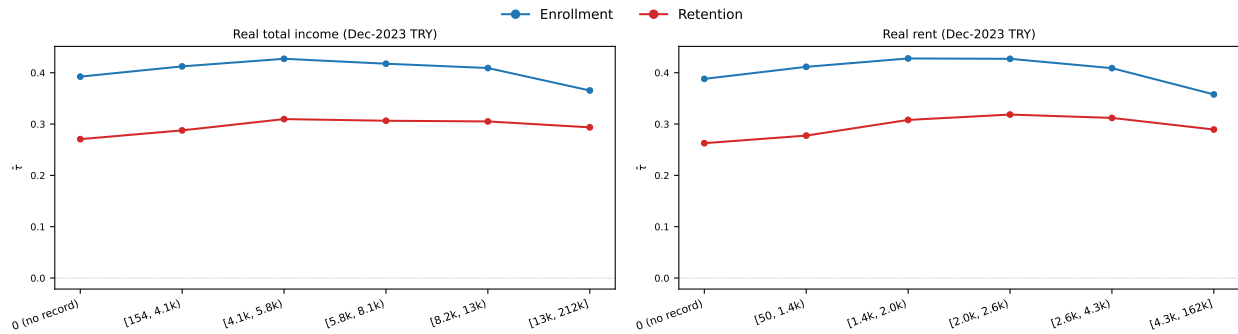


Figure EC.6 Marginal CATE on economic moderators: real total household income and real monthly rent, both CPI-deflated to December-2023 TRY using the World-Bank Turkey CPI series. Each panel shows the 0-bin (no recorded value) plus five quintiles of the positive sub-distribution; bin labels report the empirical TRY range. The pattern is mildly U-shaped on both outcomes — the program’s lift is largest at the bottom and the top of the income distribution and dips slightly in the middle — with overall spreads of ≈ 5 –6 pp (Table EC.6).

proximate stakeholder (the household, the child, the medical system, or the broader crisis context). We use the unified-vocabulary labels from Appendix EC.6.1.

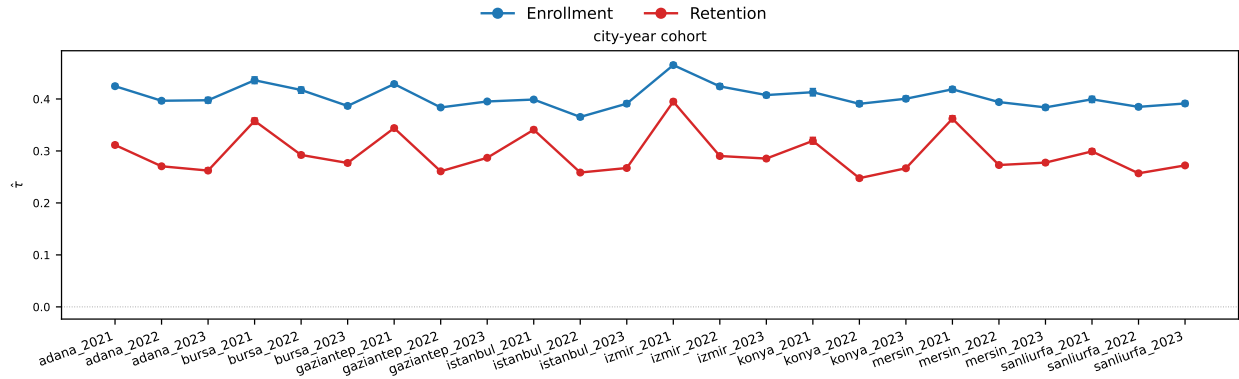


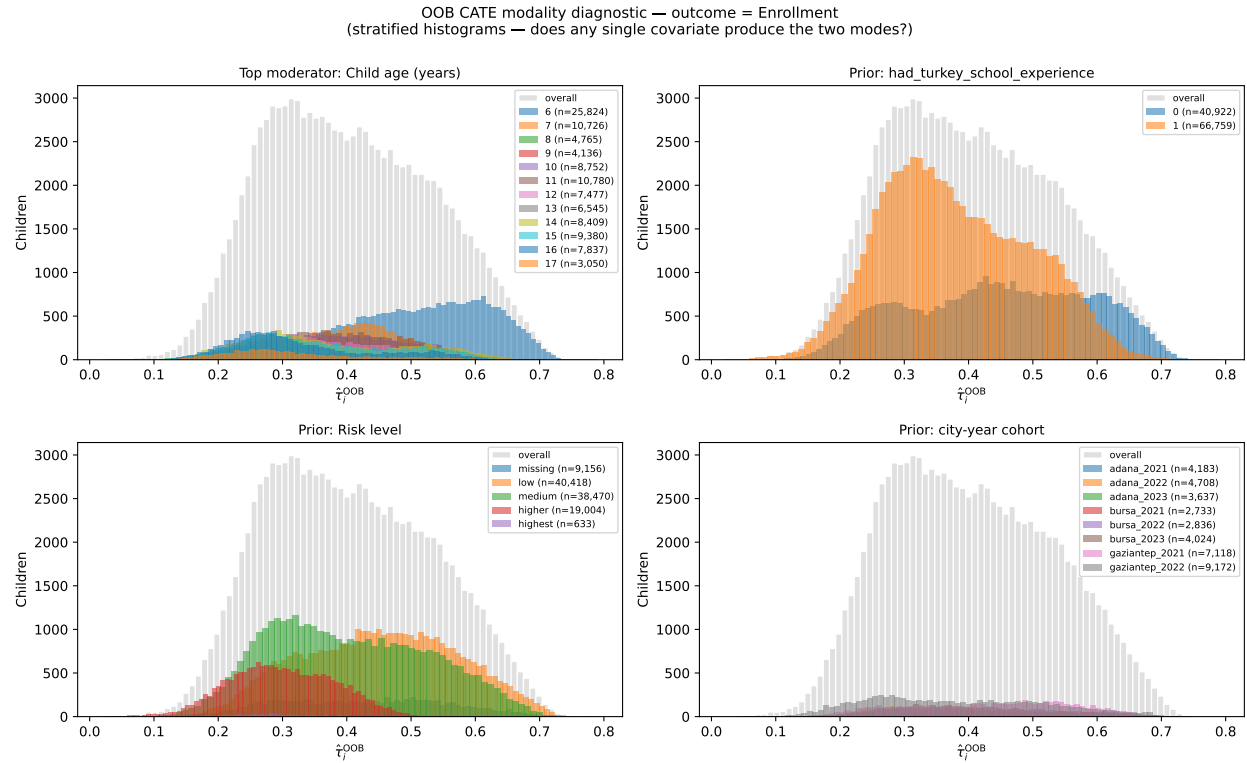
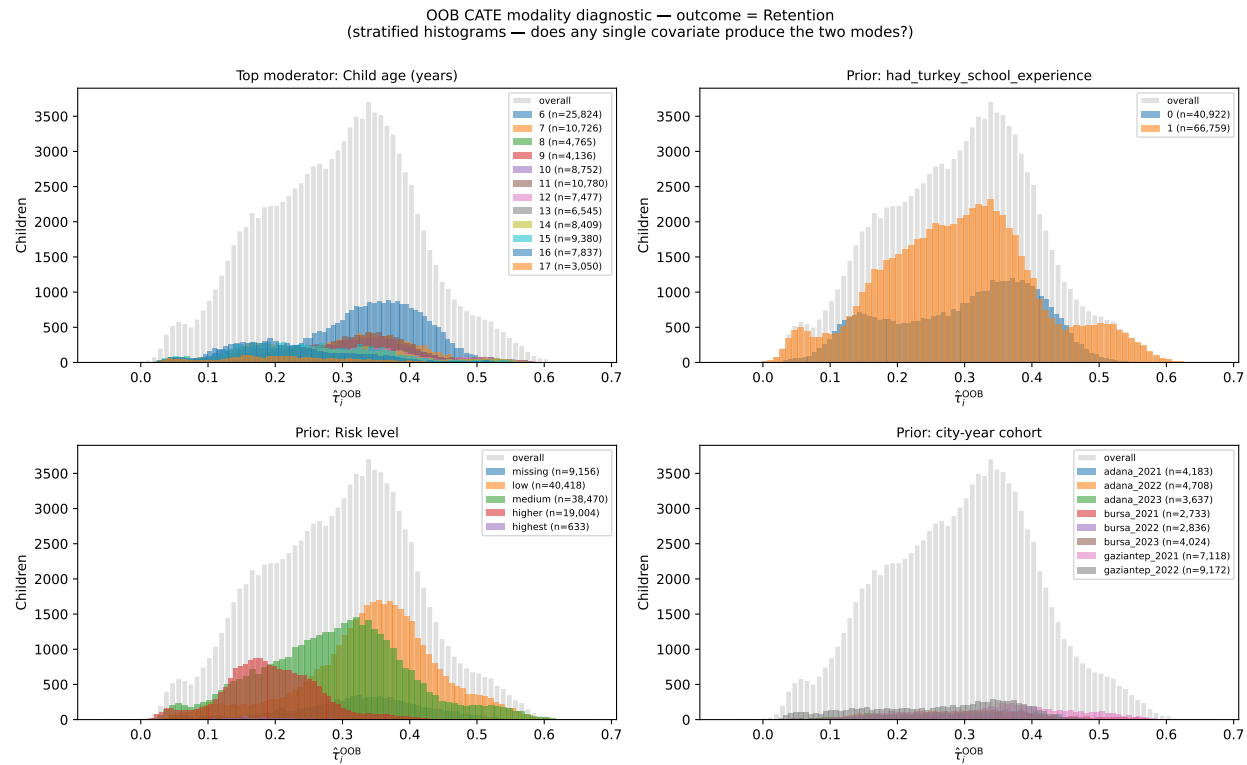
Figure EC.7 Marginal CATE by city-year cohort. The 24 cohort dummies (eight cities $\times$ three years) are the unit of identification, not a substantive moderator; their CATE differences are absorbed by the cohort fixed effects in the nuisance and reported here only as a diagnostic of cross-cohort heterogeneity. The cohort-level spread is the largest of any moderator in Table EC.6, which is by construction: cohort effects pool all city-level institutional differences, year-level policy changes, and aggregate shocks (COVID disruption, the February 2023 earthquake) that the nuisance is designed to absorb.

Table EC.7 Mapping of `barrier_X` flags to substantive barrier categories used in the row grouping of Figure 7.

Category	Member <code>barrier_X</code> flags
Structural / operational	<code>barrier_SchoolAccessLocation</code> , <code>barrier_DocumentationIssues</code>
Economic	<code>barrier_FinancialHardship</code>
Child-side	<code>barrier_ChildLabour</code> , <code>barrier_ChildRefusesSchool</code> , <code>barrier_BullyingPeer</code> , <code>barrier_LanguageBarrier</code> , <code>barrier_CulturalDiscrim</code>
Family-side	<code>barrier_NoFamilySupport</code> , <code>barrier_GenderForcedMarriage</code>
Health / disability	<code>barrier_HealthIssues</code> , <code>barrier_Disability</code>
Crisis / displacement	<code>barrier_EarthquakeOrCrisis</code> , <code>barrier_RelocationDisplacement</code>

EC.4.6. OOB CATE modality diagnostic

The bimodal OOB CATE distribution noted in subsection 5.2.1 prompted a single-covariate decomposition: for every summary moderator, what fraction of the variance in $\hat{\tau}$ is explained by level membership? The metric is the categorical-ANOVA $\eta^2 = \text{SSB}/\text{SST}$. A covariate that creates two well-separated modes drives η^2 up because most of the $\hat{\tau}$ variance is between-level, not within-level. Figure EC.8 and Figure EC.9 report stratified histograms for the top moderator (by $\eta^2_{\text{retention}}$, namely `barrier_SchoolAccessLocation`), the prior moderator we initially tested (`had_turkey_school_experience`), risk level, and city-year cohort. For retention the school-access barrier is the cleanest single-covariate explanation: each level's sub-distribution is approximately unimodal and the two modes line up with the marginal modes ($\eta^2 = 0.28$). For enrollment no single moderator dominates ($\eta^2 \leq 0.28$ across all candidates), and the bimodality is multi-causal.

**Figure EC.8** Modality diagnostic for the enrollment OOB CATE.**Figure EC.9** Modality diagnostic for the retention OOB CATE.

EC.5. Robustness Appendix

EC.5.1. Within-cohort placebo permutation

We test the null hypothesis that, conditional on the city-year cohort structure, timely support has no effect on the residualized outcome. Under this null, the Robinson estimand $\hat{\theta}_0$ has expectation zero and any sign of a shift of the observed estimate from zero is informative. Because $\tilde{D}_i$ and $\tilde{Y}_i$ are already centered by the cross-fitted nuisance, a valid placebo is to permute $\tilde{D}_i$ *within* each city-year cohort (preserving the within-cohort marginal distribution) and recompute $\hat{\theta}_0$ against the untouched $\tilde{Y}_i$. Table EC.8 reports the observed estimate alongside the placebo mean, SD, and two-sided tail probability across $B = 500$ permutations; Figure EC.10 visualizes the placebo distributions. For both outcomes, the observed $\hat{\theta}_0$ lies well beyond the entire placebo distribution.

Table EC.8 Within-cohort placebo permutation test (500 draws). Under the null that timely support has no effect once the cohort structure is controlled, the residualised Robinson estimand $\hat{\theta}_0$ has mean zero. We reshuffle $\tilde{D}$ within each city-year cohort and recompute $\hat{\theta}_0$ against the un-permuted $\tilde{Y}$. The observed estimate lies in the extreme tail of the placebo distribution for both outcomes.

Outcome	Observed $\hat{\theta}_0$	Placebo mean	Placebo SD	Two-sided p
Enroll	0.3817	-0.00001	0.00398	< 0.002
Retention	0.2609	+0.00001	0.00468	< 0.002

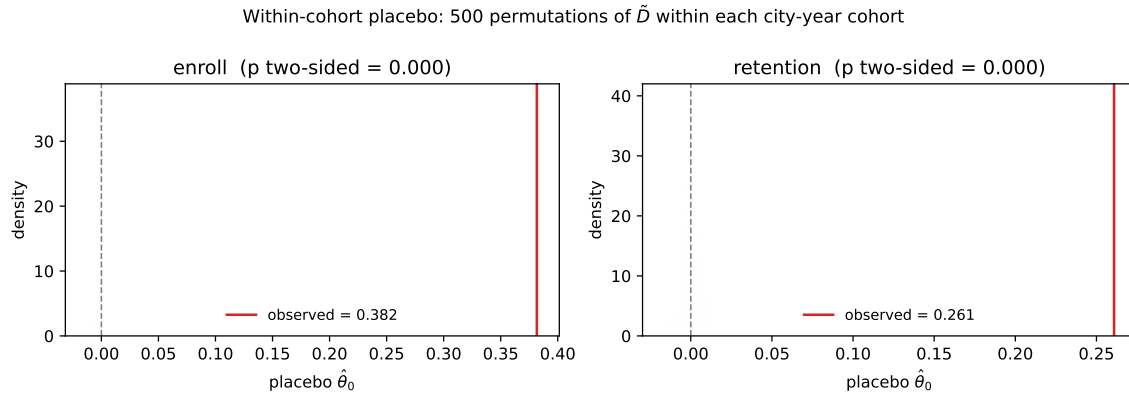


Figure EC.10 Within-cohort placebo distribution of $\hat{\theta}_0$ from $B = 500$ permutations of $\tilde{D}_i$ within each city-year cohort. Red line: observed estimate.

EC.5.2. Alternative specifications

Table EC.9 reports OLS (linear probability), logit, and probit estimates of the treatment effect using the same design matrix that feeds the DML nuisance step. All three parametric models give estimates noticeably above DML for both outcomes. The gap is consistent with the L1-logit

DML result in Table EC.10: parametric nuisance models cannot absorb the nonlinear interactions between cohort-level baseline rates, age, and parental structure that tree ensembles pick up. The DML-ATE is thus not an artifact of the flexible first stage; it is the more conservative estimate.

Table EC.9 Alternative specifications for the average treatment effect. Each panel-row reports the estimated average partial effect of timely enrollment support from a different parametric specification; the DML-ATE row at the top reproduces the headline estimate from Table EC.3. All models include the full covariate set X and city-year cohort fixed effects. Family-clustered standard errors in parentheses.

Specification	Enrollment	Retention
DML-ATE (LightGBM)	0.3817 (0.0050)	0.2609 (0.0039)
OLS (LPM)	0.4241 (0.0047)	0.3149 (0.0038)
Logit (APE)	0.4493 (0.0063)	0.3833 (0.0043)
Probit (APE)	0.4401 (0.0070)	0.3738 (0.0046)

EC.5.3. First-stage learner sensitivity

Table EC.10 repeats the DML-ATE across all four nuisance learners. The three nonparametric learners cluster within one percentage point of each other on both outcomes; the L1 logit shifts upward by roughly six to seven percentage points, mirroring the shift produced by the parametric alternative specifications in Appendix EC.5.2. Read alongside the nuisance AUC in Table EC.1 — LightGBM / XGBoost / Random Forest at 0.80–0.82 versus L1 logit at 0.75–0.78 — the pattern identifies linear nuisance bias rather than forest-specific artifacts as the source of the divergence.

Table EC.10 Sensitivity of the DML-ATE to the choice of first-stage learner. Each column varies only the learner used to estimate the nuisance functions $m(X, C)$ and $g(X, C)$; the residual-on-residual Robinson regression is identical across columns. Family-clustered standard errors in parentheses; two-way (family $\times$ city-year) clustered standard errors in brackets. The three nonparametric learners agree to within one percentage point; the L1-logit is inflated relative to them, consistent with linear nuisance functions leaving confounding interactions unabsorbed.

Outcome	LightGBM	XGBoost	Random Forest	L1-logit
Enroll	0.3817 (0.0050) [0.0371]	0.3823 (0.0050) [0.0373]	0.4234 (0.0046) [0.0368]	0.4263 (0.0048) [0.0332]
Retention	0.2609 (0.0039) [0.0287]	0.2612 (0.0040) [0.0290]	0.3063 (0.0035) [0.0301]	0.3126 (0.0038) [0.0264]

EC.5.4. Earthquake impact

The February 2023 earthquake affected Turkey’s southeast for much of the 2023 cohort’s outcome-observation window. Table EC.11 presents two sensitivity exercises. Panel A augments the Robinson regression with a post-earthquake treatment interaction (indicator for assessment on or after 6 February 2023): the interaction is a precisely estimated +2.1 percentage points on enrollment and −2.9 points on retention. The sign pattern is intuitive — earthquake-era assessments did not lose their enrollment-facilitation value, but post-enrollment retention in the affected areas was lower. Panel B excludes the three southeastern cities in our sample (Adana, Gaziantep, Sanliurfa), reducing the sample to 67,071 children; the resulting $\hat{\theta}_0 = 0.51/0.30$ sits above the full-sample DML-ATE of 0.47/0.27, suggesting the earthquake-affected cohorts pulled the pooled estimate modestly downward on both margins.

Table EC.11 Sensitivity of the DML-ATE to the February 2023 Turkey earthquake. Panel (A) interacts timely support with a post-earthquake indicator (assessment date on or after 6 February 2023); the interaction coefficient captures the differential treatment effect for assessments made after the earthquake. Panel (B) re-estimates the baseline DML-ATE excluding the southeastern cities affected by the earthquake.

Family-clustered standard errors in parentheses.		
	Enrollment	Retention
<i>Panel A: post-earthquake interaction</i>		
$\hat{\theta}_0$ (baseline)	0.3755 (0.0060)	0.2646 (0.0048)
$\hat{\theta}_{\text{post-eq}}$ (interaction)	+0.0176 (0.0106)	-0.0108 (0.0083)
$N_{\text{post-eq}}$	32,483	
<i>Panel B: excluding southeastern cities</i>		
$\hat{\theta}_0$ (subsample)	0.4279 (0.0063)	0.2915 (0.0052)
N	59,717 (dropped 47,964)	
DML-ATE reference	0.3817 (0.0050)	0.2609 (0.0039)

EC.5.5. Within-family spillovers

Because the sample contains 58,437 families with multiple children, within-household information spillover is the first-order SUTVA concern: a child whose sibling was treated has indirect exposure to the program (through the parents). Table EC.12 partitions the sample into three strata — treated (T), clean control (S_0 : control with no sibling treated), and spillover-exposed control (S_1 : control with at least one sibling treated) — and reports both raw outcome rates and two augmented Robinson regressions. Panel A augments the second stage with an S_1 indicator on the full

sample; the point estimate of $\hat{\theta}_0$ is essentially unchanged from the headline estimate (0.472/0.263 vs. 0.474/0.265), and the sibling-exposure coefficient is a precisely estimated +1 percentage point on both outcomes. Panel B restricts the sample to the control group and regresses $\tilde{Y}_i$ on the S_1 indicator alone; the coefficient is statistically indistinguishable from zero (and in fact negative), indicating that after conditioning on X and C the S_1 stratum does not differ from S_0 on the outcome margin. The two panels together support the view that within-family spillovers are small relative to the direct effect.

Table EC.12 Within-family spillover diagnostic. Children are sorted into three strata based on their own treatment status and that of their siblings within the same family: T (treated); S_0 (control with no sibling treated); S_1 (control with at least one sibling treated). Panel (top) gives raw enrollment and retention rates by stratum. Panel A augments the Robinson second stage with an S_1 indicator; β_{S_1} captures the extra outcome variation attributable to having a treated sibling. Panel B restricts the sample to the control group and regresses the residualized outcome on S_1 alone. Family-clustered standard errors in parentheses.

	Enrollment	Retention
<i>Raw outcome rates by stratum</i>		
Treated ($n = 95,241$)	0.7225	0.4960
S_0 (control, no sib treated) ($n = 7,190$)	0.1513	0.0423
S_1 (control, sib treated) ($n = 5,250$)	0.1779	0.0781
<i>Panel A: augmented Robinson regression (full sample)</i>		
$\hat{\theta}_0$	0.3793 (0.0050)	0.2572 (0.0039)
β_{S_1}	+0.0109 (0.0027)	+0.0170 (0.0030)
<i>Panel B: control-only regression</i>		
β_{S_1}	-0.0237 (0.0079)	-0.0063 (0.0057)

EC.5.6. Cohort-covariate R^2 diagnostic

subsubsection 4.2.2 argues that because the policy-targeting problem allocates within a given city-year cohort, cohort identity should enter the nuisance but not the heterogeneity surface. This design is valid only if individual covariates are not collinear with cohort membership. Table EC.13 reports the R^2 of regressing each of the 74 individual covariates on the 24 city-year cohort dummies. The median R^2 is 0.03; the 99th percentile is 0.13; the maximum is 0.14. The fifteen covariates most strongly absorbed by cohort (accommodation type, Syrian ethnicity, mother Turkish proficiency, parental employment) are reported in the table. Because no covariate exceeds $R^2 = 0.20$, cohort FEs cannot drain the identifying variation out of the individual covariates used in the heterogeneity surface.

Table EC.13 Cohort-covariate R^2 diagnostic. For each individual covariate x_j we regress x_j on the 24 city-year cohort dummies and report the resulting R^2 (continuous x_j) or McFadden pseudo- R^2 (binary x_j). The fifteen covariates most strongly absorbed by cohort are shown; the remaining 174 covariates have $R^2 \leq 0.122$. Median $R^2 = 0.039$.

Covariate	Type	R^2
Dropout_NoEBAAccess	binary	0.381
barrier_EarthquakeOrCrisis	binary	0.347
Dropout_Earthquake	binary	0.340
need_cat_persons_affected_by_disaster	binary	0.317
need_sn_affected_by_earthquake	binary	0.317
NotAttend_Earthquake	binary	0.193
TypeOfCase_calling_assessment	binary	0.164
need_sn_unmet_basic_needs	binary	0.158
need_cat_specific_legal_and_physical_protection_n	binary	0.157
need_cat_victim_of_human_trafficking	binary	0.155
accom_formal_detached	binary	0.138
rent_quintile	continuous	0.138
accom_formal_squatter	binary	0.134
income_quintile	continuous	0.125
need_sn_lack_of_durable_solutions_prospects	binary	0.122

EC.5.7. Covariate balance between treated and control children

Our identification assumption requires treated and control children to be comparable on pre-treatment covariates *conditional* on the cohort structure and the observable priority rule. Standardized mean differences (SMDs) are the conventional diagnostic for this kind of balance in observational studies: unlike a t -statistic they are sample-size-free and can be compared across covariates of different scales (Rosenbaum and Rubin 1985, Stuart 2010, Austin 2009). For a binary indicator or continuous covariate X_k , we define within-cell SMD as

$$d_k^{(c,s)} = \frac{\bar{X}_{k,T}^{(c,s)} - \bar{X}_{k,C}^{(c,s)}}{\sqrt{[(s_{k,T}^{(c,s)})^2 + (s_{k,C}^{(c,s)})^2]/2}}, \quad (\text{EC.1})$$

where $\bar{X}_g^{(c,s)}$ and $s_g^{(c,s)}$ denote the mean and standard deviation of X_k in treatment group $g \in \{T, C\}$ restricted to cell (c, s) . The cell (c, s) indexes a city-year cohort c and a within-cohort stratum s . Table EC.14 reports three complementary summaries of (EC.1):

(i) *Unconditional SMD, cohort-weighted.* Setting s to a single dummy stratum (i.e., no within-cohort stratification) yields the within-cohort unconditional SMD $d_k^{(c)}$; we then pool across cohorts with sample-size weights,

$$d_k^{\text{unc}} = \frac{\sum_c n^{(c)} d_k^{(c)}}{\sum_c n^{(c)}}, \quad n^{(c)} = n_T^{(c)} + n_C^{(c)}.$$

This is the diagnostic one would report if treated and control children were simply compared within cohort, without conditioning on the priority rule. Because cohort fixed effects already enter the first-stage nuisance, we view d_k^{unc} as the *raw* imbalance that the DML procedure must then absorb.

(ii) *Conditional (stratum-pooled) SMD, cohort-weighted.* The program’s priority rule ranks children primarily on Turkish school experience and age, so in-control children who look like the treated on these two dimensions are the relevant counterfactual. We therefore stratify each cohort on the joint variable $\tilde{S}_i = (\text{had_turkey_school_experience}_i, \text{age_bin}_i)$ — two school-experience groups crossed with 12 one-year age bins, centred on integer ages 6–17 (bin a collects children with $\text{age_at_assess} \in [a - 0.5, a + 0.5)$; the sample is restricted to children assessed at age ≥ 5.5 , so the youngest bin is age 6 = $[5.5, 6.5)$). The pooled conditional SMD within a cohort, weighted by cell size, is

$$d_k^{(c),\text{cond}} = \frac{\sum_s n^{(c,s)} d_k^{(c,s)}}{\sum_s n^{(c,s)}},$$

which is the Mantel–Haenszel-style stratum pool used in Rosenbaum and Rubin (1985) and recommended in Stuart (2010) and Ho et al. (2007) as the primary diagnostic when the target conditioning set is known. Pooling again across cohorts,

$$d_k^{\text{cond}} = \frac{\sum_c n^{(c)} d_k^{(c),\text{cond}}}{\sum_c n^{(c)}}.$$

A small d_k^{cond} means that once we condition on cohort, school experience, and age — the dimensions that the program’s operational priority rule selects on — the treated and control distributions of X_k are essentially indistinguishable.

(iii) *Share of well-balanced cells.* A pooled mean can mask imbalance that cancels across cells. We therefore also report, for each covariate, the share of (cohort $\times \tilde{S}$) cells in which $|d_k^{(c,s)}| < 0.25$ — the threshold Imbens and Rubin (2015) flag as “substantial” imbalance and which corresponds to a Cohen- d of 0.25, around the boundary Stuart (2010) cites between acceptable and concerning imbalance (other empirical studies use the stricter 0.10 of Austin (2009); we report the looser 0.25 since our stratifier already controls the most imbalanced covariates by construction). The percentage is computed only over cells with at least two treated and two control children, so sparsely populated cells do not inflate the denominator.

Reading the table. The stratifier `had_turkey_school_experience` and the 12 age-bin dummies are mechanically balanced once we condition on $\tilde{S}$ — $d_k^{\text{cond}} = 0$ and 100% of cells are balanced — which is a sanity check on the stratification rather than a finding. The informative rows are the remaining 54 covariates: the largest unconditional gap, $d_k^{\text{unc}} = 0.48$ on `had_turkey_school_experience` itself, is absorbed entirely by the joint stratifier, and no other covariate has $|d_k^{\text{cond}}| > 0.10$. The five largest residual gaps are on Turkish-language proficiency variables (child’s spoken Turkish, father’s and mother’s Turkish self-reports; each with $|d_k^{\text{cond}}|$ between 0.07 and 0.10), which is consistent with the priority rule giving weight to Turkish school experience — through which Turkish language exposure and family tenure partially correlate — but

no explicit weight to language per se. Every covariate has $|d_k^{\text{cond}}| < 0.10$ once we condition on $\tilde{S}$, and the share of cells balanced at the 0.25 threshold is above 38% for every covariate. Because the DML nuisance absorbs all 74 individual covariates nonparametrically, any residual imbalance these secondary variables capture is differenced out in the Robinson second stage; the conditional SMD in column (ii) is best read as a descriptive statement of how much of the imbalance would have survived a purely stratified analysis.

Table EC.14: Cohort-conditional covariate balance between treated and control children. Columns report, for each baseline covariate: (i) the cohort-weighted unconditional standardized mean difference d_k^{unc} ; (ii) the cohort-weighted, stratum-pooled conditional standardized mean difference d_k^{cond} , where the within-cohort stratifier is $\tilde{S}_i = (\text{had_turkey_school_experience}_i, \text{age_bin}_i)$ with 13 age bins (ages 5–17) and two school-experience strata, giving 26 cells per cohort; (iii) the fraction of $(\text{cohort} \times \tilde{S})$ cells in which $|d_k| < 0.25$, the Imbens–Rubin large-imbalance threshold. The risk-assessment and schooling-barrier panels add summary covariates derived from the programs assessment instruments. The final condensed panel collapses the high-dimensional text-encoded controls (specific-needs and per-status barrier dummies) into one summary row per group; these covariates enter the DML/CRF nuisance learners but are not interpreted individually.

Variable	d_k^{unc}	d_k^{cond}	% Balanced cells ($ d < 0.25$)
<i>Child Demographics, Family, Accommodation</i>			
had_turkey_school_experience	+0.340	+0.000	
age_11	+0.181	+0.000	
age_10	+0.137	+0.000	
age_16	-0.113	+0.000	
accom_formal_flat	+0.089	+0.059	
family_two_parents	+0.069	+0.016	
age_15	-0.068	+0.000	
age_17	-0.067	+0.000	
turkish_medium	+0.066	+0.069	
accom_informal	-0.065	-0.016	
has_siblings	+0.058	-0.028	
age_6	-0.048	+0.000	
age_12	+0.047	+0.000	
is_syrian	+0.046	-0.035	
is_eldest	-0.046	+0.016	
turkish_high	+0.045	+0.054	
is_female	+0.042	+0.048	
age_7	+0.037	+0.000	
turkish_low	+0.036	+0.045	
shared_house	-0.036	+0.017	
is_single	+0.035	-0.020	
is_youngest	-0.034	+0.002	
family_size	+0.034	+0.019	
family_single_father	-0.031	+0.017	
accom_formal_detached	-0.031	-0.006	
age_14	-0.027	+0.000	
age_13	-0.026	+0.000	
household_size	+0.025	+0.033	
family_single_mother	-0.020	+0.030	
real_rent	+0.015	+0.033	

(continued)

Variable	d_k^{unc}	d_k^{cond}	% Balanced cells ($ d <$
accom_formal_other	-0.013	+0.019	
age_9	-0.011	+0.000	
real_income	+0.009	+0.023	
grandparent_present	+0.008	+0.038	
accom_formal_squatter	+0.004	+0.034	
turkish_native	+0.001	+0.010	
age_8	+0.001	+0.000	
<i>Risk-Assessment Summary</i>			
TypeOfCase_complex	-0.390	-0.322	
RiskLevel_higher	-0.384	-0.322	
TypeOfCase_intermediate	+0.352	+0.293	
need_cat_child_at_risk	-0.215	-0.176	
need_cat_disability	-0.191	-0.073	
RiskLevel_highest	-0.115	-0.031	
need_cat_serious_medical_condition	-0.100	-0.039	
need_cat_unaccompanied_or_separated_child	-0.095	-0.012	
need_cat_gbv	-0.084	-0.006	
need_cat_persons_affected_by_disaster	-0.075	-0.076	
RiskLevel_medium	+0.071	+0.101	
need_cat_specific_legal_and_physical_protection_n	-0.070	-0.083	
is_miss_RiskLevel	-0.012	+0.034	
is_miss_TypeOfCase	-0.012	+0.034	
need_cat_covid_19	+0.011	+0.004	
need_cat_family_unity	-0.010	+0.004	
TypeOfCase_initial	+0.007	+0.033	
need_cat_parent_or_caregiver_with_specific_needs	+0.005	+0.002	
need_cat_alleged_perpetrator	+0.004	+0.002	
need_cat_victim_of_human_trafficking	-0.003	+0.000	
need_cat_woman_at_risk	-0.002	+0.000	
need_cat_torture	+0.001	+0.015	
TypeOfCase_calling_assessment	+0.001	+0.042	
<i>Schooling-Barrier Summary</i>			
barrier_SchoolAccessLocation	+0.575	+0.396	
barrier_DocumentationIssues	-0.281	-0.098	
barrier_Disability	-0.221	-0.087	
barrier_HealthIssues	-0.184	-0.117	
barrier_ChildRefusesSchool	-0.133	-0.055	
barrier_RelocationDisplacement	-0.117	-0.111	
barrier_ChildLabour	-0.104	-0.022	
n_barriers_unique	-0.100	-0.105	
barrier_GenderForcedMarriage	-0.086	-0.003	
barrier_NoFamilySupport	-0.084	-0.004	
barrier_NotSchoolAge	-0.052	+0.025	
barrier_EarthquakeOrCrisis	-0.049	-0.075	
barrier_FinancialHardship	-0.046	+0.013	
barrier_LanguageBarrier	-0.031	+0.024	
any_barrier_reported	+0.025	-0.021	
barrier_BullyingPeer	-0.024	-0.009	
barrier_CulturalDiscrim	-0.022	+0.011	
barrier_SeasonalAgriWork	+0.000	+0.000	
<i>High-dim text-encoded controls (condensed; one row per group)</i>			

(continued)

Variable	d_k^{unc}	d_k^{cond}	% Balanced cells ($ d <$
Parental characteristics (presence-interacted dummies + ages) (30 cols), mean $ d $	0.030	0.042	
max $ d $	0.077	0.102	
cols with $ d^{\text{unc}} > 0.10$	0	–	
need_sn_* (specific-needs detail) (27 cols), mean $ d $	0.082	0.034	
max $ d $	0.294	0.189	
cols with $ d^{\text{unc}} > 0.10$	7	–	
Per-status barrier detail (Dropout/IrregAttend/NotAttend) (37 cols), mean $ d $	0.075	0.040	
max $ d $	0.511	0.332	
cols with $ d^{\text{unc}} > 0.10$	7	–	

EC.6. Variable Definitions and Feature Engineering

This appendix documents all variables used in the estimation. Table EC.15 provides the complete list, organized by category. Our sample covers three program years (2021–2023) and eight cities, yielding 24 city–year cohorts used as fixed effects.

EC.6.1. Variable Construction

Sample filtering. We drop observations with unreasonable or structurally missing core demographics rather than imputing them, because such cases typically reflect data-entry issues rather than real variation: (i) *age_at_assess* missing or ≥ 18 (the program targets children under 18); (ii) *is_female* missing (no sensible default); (iii) *household_size* missing after the family-size fallback described below. Combined, these drop rules remove about 1% of the raw first-entry sample. We retain *is_single* (set to 1 when missing, reflecting that almost all children under 18 are single). Family economic fields (*TotalIncome*, *Rent*) are deflated to December-2023 TRY using the World-Bank Turkey CPI series (FP.CPI.TOTL) at the family’s *RegisterDate*, and the deflated values are stored as *real_income* and *real_rent*; families with no recorded value receive 0 and are flagged through the 0-bin of the derived quintile variables *income_quintile* and *rent_quintile* (see “Economic moderators” below).

Age bins. After filtering to children under 18, we discretize child age at assessment into one-year bins: [5, 6) (reference), [6, 7), $\dots$, [17, 18), yielding 12 dummy variables. This avoids inflated feature importance in tree-based CATE estimators relative to a single continuous age variable.

Turkish language proficiency. We create four indicator variables from the categorical language field: *low*, *medium*, *high*, and *native*. The reference category is “don’t know.”

Family structure. We characterize the child’s household along two dimensions. First, three *nuclear-family* dummies indicate parental composition (*two parents*, *single mother*, *single father*), with reference category orphan (no parent linked to the child via the parent-link table). Second, three extended-household indicators record whether the child lives with *grandparents* (*grandparent_present*), *has siblings* (*has_siblings* = 1 iff *n_siblings* > 0), and is the oldest child in the family (*is_eldest*). *is_eldest* equals 1 when the focal child has the earliest date of birth among all household members born on or after 2003-01-01 (ties count all tied children as eldest).

Grandparent identification. The raw data records each person’s **Relative** relationship to the household’s Primary Applicant (PA), not directly to the focal child. We therefore construct *grandparent_present* by conditioning on the child’s own **Relative** value: (i) if the child is *Son/Daughter (or Step Child) of PA*, a grandparent is present when any other household member has **Relative** in {Father of PA, Mother of PA, Father-in-Law of PA, Mother-in-Law of PA, Grand Parent of PA}; (ii) if the child is *Grand Child of PA*, *grandparent_present* equals 1 by construction (the PA is the

grandparent); (iii) if the child is the *PA* themselves, a grandparent is present only if **Grand Parent of PA** is in the household (“Father/Mother of PA” would be the PA’s parents, not grandparents); (iv) for all other kin categories (Blood-kin, Non-blood kin, Brother/Sister of PA, step-kin, etc.), the generation relative to the child is ambiguous and we default to 0.

Accommodation type. Five dummies capture housing: *formal flat*, *formal detached*, *formal squatter*, *informal*, and *formal other*. The reference is missing/unrecorded accommodation type.

Parental characteristics. All parent-level variables (age, language, marital status, employment, education) are interacted with a parent-presence indicator. If the father (mother) is absent, all father (mother) characteristics are set to zero. This avoids conflating missing-parent with missing-data.

We simplify the raw parental attributes along two dimensions to keep the control set interpretable. First, parental marital status collapses to two dummies: *marital_alone* (single, widowed, divorced, or separated) and *marital_missing*, with reference category *married/in-relationship*. Second, parental education collapses the 15-level raw scale into three dummies: *edu_none* (never attended school, including kindergarten and special education), *edu_advanced* (vocational, bachelors, or masters), and *edu_missing*, with reference category *edu_basic* (primary, middle, or high school). This grouping reflects Turkey’s 12-year compulsory schooling policy, under which primary, middle, and high school are all part of compulsory education.

Cohort fixed effects. We define a cohort as a unique city–year pair. With 8 cities (Gaziantep, Istanbul, Izmir, Adana, Şanlıurfa, Mersin, Bursa, Konya) and 3 years (2021–2023), we have 24 cohorts. Gaziantep–2021 serves as the reference, yielding 23 cohort fixed-effect dummies that absorb city-level and year-level unobserved heterogeneity.

Risk-Assessment covariates. Each refugee family that ASAM intakes also receives a separate, structured *risk assessment* that produces (i) a categorical **RiskLevel** in the ordered set $\{low, medium, higher, highest, no\ assessment\}$ and (ii) a multi-select list of *specific needs* drawn from a four-layer ASAM hierarchy (Category $\rightarrow$ SpecificNeeds $\rightarrow$ SpecificNeeds1 $\rightarrow$ SpecificNeeds2; the full hierarchy and node-level counts are reported in the auxiliary file `paper/tab/risk_hierarchy_full.csv`). For each focal child–year observation, we link to the risk assessment whose **AssessmentDate** is closest to the child’s first education- assessment date (**first_assess_date**), within a ± 365 -day window. Two source values are treated as missing rather than as their own category: **RiskLevel** $\in \{no\ assessment, not\ completed\}$ and **TypeOfCase** $\in \{initial\ no\ assessment, not\ completed\}$. To minimise the residual missing rate, we use a two-stage matching procedure: (i) closest *valid* assessment in the ± 365 -day window; (ii) if no valid assessment falls in the window, fall back to the closest valid assessment for the same **MemberID** anywhere in the table. When even the fallback returns no valid record, we set `is_miss_RiskLevel` = 1 and code

all RiskLevel/TypeOfCase dummies to zero. This procedure yields a 91.5% risk-match rate at the child–year level (versus 72.7% under the single-stage in-window match).

We construct three groups of risk-assessment covariates:

- *RiskLevel dummies* (3 active levels with reference *low*): *RiskLevel_medium*, *RiskLevel_higher*, *RiskLevel_highest*, plus the missingness flag *is_miss_RiskLevel*. *Low* is the modal valid level in matched assessments and serves as a clean ordered baseline against the higher-severity contrasts. The program’s residual codes (*no assessment*, *not completed*) are absorbed into the missing bucket via the two-stage matching procedure above.

- *TypeOfCase dummies* (4 levels with reference *missing*): *TypeOfCase_intermediate*, *TypeOfCase_complex*, *TypeOfCase_calling_assessment*, *TypeOfCase_initial*, plus the missingness flag *is_miss_TypeOfCase*. The residual *initial no assessment* and *not completed* codes are treated as missing rather than getting their own dummies.

- *Need-category dummies* (14 levels, one per top-level Category in the ASAM hierarchy, e.g., *Child at Risk*, *Specific legal and physical protection needs*, *Persons Affected by Disaster*, *Disability*, *Serious medical condition*, *GBV*, *Unaccompanied or separated child*, ...). All flags are zero for unmatched rows. (Earlier revisions also emitted the count summaries *n_specific_needs* and *n_need_categories*; the 2026-04-30 revision drops them as redundant with the deduplicated barrier-diversity count *n_barriers_unique* below.)

Schooling-barrier covariates (summary). The education-assessment instrument also captures the family’s self-reported *barriers to schooling* via three multi-select questions conditional on the child’s status (drop out, irregular attendance, never attended). Each selected option is recorded as a separate row in **EducationAssessmentDetails**; we translate the raw Turkish free-text answers using the same controlled vocabulary applied in Figure 1 (16 unified barrier categories) and emit:

- Two summary indicators: the catch-all *any_barrier_reported* (1 if any non-Other barrier flagged in any of the three barrier questions) and the deduplicated barrier-diversity count *n_barriers_unique* (the number of *distinct* unified-barrier topics raised; constructed as the sum of the unified-barrier dummies described next, so that the same issue cited in two of the three branches is counted once).

- Sixteen *unified-barrier* dummies of the form *barrier_X* (e.g., *barrier_FinancialHardship*, *barrier_LanguageBarrier*, *barrier_SchoolAccessLocation*, *barrier_ChildLabour*, *barrier_Disability*, ...), each set to 1 if the child cited *any* source token mapped to that unified label across the three barrier questions. (Earlier revisions also emitted the per-branch counts *n_dropout_reasons*, *n_irregattend_reasons*, *n_notattend_reasons* and their sum *n_barriers_total*; the 2026-04-30 revision drops them as collinear with *n_barriers_unique* once the deduplication step is applied.)

The summary covariates above appear in the main covariate-balance and summary-statistics tables (Table EC.14, Table EC.16).

High-dimensional text-encoded controls. In addition to the summary covariates above, we feed two sparse blocks of text-encoded indicators into the DML/CRF nuisance learners: (i) *need_sn_X* dummies for each **SpecificNeeds** token with $\geq 1,000$ occurrences in the source table (yielding ~ 27 dummies, e.g., *need_sn_unmet_basic_needs*, *need_sn_affected_by_earthquake*, *need_sn_violence_abuse_or_neglect_children*, *need_sn_critical_medical*); and (ii) per-status barrier dummies of the form *Dropout_X*, *IrregAttend_X*, *NotAttend_X* (yielding ~ 37 non-empty columns after dropping all-zero columns). These two blocks together comprise ~ 64 high-dimensional dummies.

The detail blocks are not listed individually in Table EC.15; they are reported in condensed form in the final panel of Table EC.16 and Table EC.14. Each detail dummy has low marginal prevalence (mean prevalence $\approx 4\%$ for need-detail, $\approx 4\%$ for barrier-detail), making them well-suited to L1-penalised or tree-based nuisance estimators. The standardised mean differences for these dummies are also small in absolute value: among the 27 specific-need dummies, mean $|d^{\text{unc}}| \approx 0.04$ and only 2 of 27 exceed the 0.10 “small imbalance” threshold; among the 37 per-status barrier dummies, mean $|d^{\text{unc}}| \approx 0.07$ with 9 of 37 exceeding 0.10.

High-dim text covariates and the choice of estimator. The volume and sparsity of these text-derived dummies provide an additional methodological motivation for the double-machine-learning (DML) and causal-forest (CRF) framework adopted in subsection 4.2–subsection 4.1. Following Chernozhukov et al. (2018a), we partial out the conditional means of the treatment and outcome using flexible regularised learners (gradient boosting and L1-logit), which are theoretically justified for sparse high-dimensional designs and remain orthogonal to the parameter of interest under cross-fitting. Linear specifications (OLS, logit, probit) cannot absorb the same set of dummies without saturating the design, but the same covariates enter our DML nuisance functions naturally and improve predictive AUC for both treatment and outcome (Appendix EC.5.3). The causal forest (Athey et al. 2019) likewise handles high-dim sparse covariates through tree-based splits, allowing the resulting CATE estimates to capture barrier-specific heterogeneity that a parametric specification would have to pre-specify by hand.

EC.6.2. Complete Variable List

Table EC.15: Variable Definitions

Variable	Type	Description
<i>Child Demographics (12 + 8 = 20 variables)</i>		
age_6 ... age_17	Binary	Age bin dummies, one-year intervals; ref: [5, 6)
is_female	Binary	Child is female (or non-binary)

<i>(continued)</i>		
Variable	Type	Description
is_syrian	Binary	Syrian nationality
is_single	Binary	Single marital status
turkish_low	Binary	Turkish proficiency: low
turkish_medium	Binary	Turkish proficiency: medium
turkish_high	Binary	Turkish proficiency: high
turkish_native	Binary	Turkish proficiency: native
had_turkey_school_experience	Binary	Prior school attendance in Turkey
<i>Family Characteristics (13 variables)</i>		
family_two_parents	Binary	Two-parent household
family_single_mother	Binary	Single-mother household
family_single_father	Binary	Single-father household
grandparent_present	Binary	At least one grandparent-generation relative in household (see construction)
has_siblings	Binary	Child has one or more siblings in the family
is_eldest	Binary	Child has the earliest date of birth among children in the family
is_youngest	Binary	Child has the latest date of birth among children in the family (singletons get $is_eldest = is_youngest = 1$)
sibling_treated_before	Binary	At least one other child in the same family has <i>first_action_date</i> strictly before this child's <i>first_assess_date</i> (pre-treatment for the focal child)
household_size	Continuous	Number of people in household
family_size	Continuous	Number of family members
n_siblings	Continuous	Number of siblings of the focal child (DOB $\geq 2003-01-01$)
shared_house	Binary	Family shares house with others
<i>Economic Moderators (4 variables)</i>		
real_income	Continuous	Total household income (TRY), CPI-deflated to December-2023 using the World-Bank Turkey series <code>FP.CPI.TOTL</code> at the family's <code>RegisterDate</code>
real_rent	Continuous	Monthly rent (TRY), deflated the same way
income_quintile	Ordinal $\{0, 1, \dots, 5\}$	Quintile of <i>real_income</i> among positive values; 0 = no recorded value
rent_quintile	Ordinal $\{0, 1, \dots, 5\}$	Quintile of <i>real_rent</i> ; 0 = no recorded value
<i>Accommodation (5 variables)</i>		
accom_formal_flat	Binary	Formal housing: flat/apartment
accom_formal_detached	Binary	Formal housing: detached house
accom_formal_squatter	Binary	Formal housing: squatter dwelling
accom_informal	Binary	Informal housing
accom_formal_other	Binary	Formal housing: other type
<i>Cohort Fixed Effects (23 variables)</i>		
cohort_{city}_{year}	Binary	City-year cohort indicator; ref: Gaziantep-2021
<i>Father Characteristics (15 variables, all $\times$ father_present)</i>		

<i>(continued)</i>		
Variable	Type	Description
father_present	Binary	Father is linked to child
father_age_x	Continuous	Father's age ($\times$ presence)
father_turkish_{low,medium,high,native,missing}_x	Binary	Father's Turkish proficiency ($\times$ presence)
father_marital_alone_x	Binary	Father single/widowed/divorced/separated ($\times$ presence); ref: married
father_marital_missing_x	Binary	Father marital status missing ($\times$ presence)
father_empl_{unemployed,not_working_age,missing}_x	Binary	Father's employment status ($\times$ presence); ref: employed
father_edu_none_x	Binary	Father never attended / kindergarten / special-ed ($\times$ presence); ref: edu.basic
father_edu_advanced_x	Binary	Father vocational / bachelor / master ($\times$ presence)
father_edu_missing_x	Binary	Father education missing ($\times$ presence)
<i>Mother Characteristics (15 variables, all $\times$ mother_present)</i>		
mother_present	Binary	Mother is linked to child
mother_age_x	Continuous	Mother's age ($\times$ presence)
mother_turkish_{low,medium,high,native,missing}_x	Binary	Mother's Turkish proficiency ($\times$ presence)
mother_marital_alone_x	Binary	Mother single/widowed/divorced/separated ($\times$ presence); ref: married
mother_marital_missing_x	Binary	Mother marital status missing ($\times$ presence)
mother_empl_{unemployed,not_working_age,missing}_x	Binary	Mother's employment status ($\times$ presence); ref: employed
mother_edu_none_x	Binary	Mother never attended / kindergarten / special-ed ($\times$ presence); ref: edu.basic
mother_edu_advanced_x	Binary	Mother vocational / bachelor / master ($\times$ presence)
mother_edu_missing_x	Binary	Mother education missing ($\times$ presence)
<i>Risk-Assessment Summary (23 variables)</i>		
RiskLevel_{medium,higher,highest}	Binary	RiskLevel dummies; ref: <i>low</i>
is_miss_RiskLevel	Binary	1 if no valid risk Assessment matched (after the two-stage in-window $\rightarrow$ any-record match)
TypeOfCase	Binary	TypeOfCase dummies; ref: missing
is_miss_TypeOfCase	Binary	TypeOfCase missing indicator (also absorbs the program's residual codes <i>initial no assessment</i> and <i>not completed</i>)
need_cat_{...} (14 dummies)	Binary	One indicator per ASAM top-level Category (e.g., <i>Child at Risk</i> , <i>Disability</i> , <i>Persons Affected by Disaster</i>); zero if no match
<i>Schooling-Barrier Summary (~ 19 variables)</i>		
any_barrier_reported	Binary	1 if at least one non-Other barrier flagged in any of the three barrier questions
n_barriers_unique	Continuous	Count of <i>distinct</i> unified-barrier topics raised (sum of the <i>barrier_*</i> unified dummies; deduplicates across drop-out, irregular-attendance and not-attending reasons that map into the same topic)

(continued)

Variable	Type	Description
barrier_{...} (16 dummies)	Binary	Unified-barrier dummies (rollup across the three per-status barrier questions; mapping mirrors Figure 1)
bcon_{...} (17 dummies; plot-only)	Binary	Substantively-curated consolidations used in the binary-forest plot of Figure 7: each <i>bcon_*</i> indicator unions one substantive barrier topic across the unified-barrier, drop-out, not-attending, and need-category sources. Not part of the design matrix (perfectly redundant with the underlying components).
<i>High-Dimensional Text Controls (condensed; ~64 columns total)</i>		
need_sn_{...}	Binary	~27 dummies for SpecificNeeds tokens with $\geq 1,000$ occurrences in the source table; enter DML/CRF nuisance only
{Dropout,IrregAttend,NotAttend}_{...}	Binary	~37 per-status barrier-detail dummies after dropping all-zero columns; enter DML/CRF nuisance only

EC.7. Summary Statistics

Table EC.16: Summary Statistics: Treated vs. Control. The first four panels list one variable per row. The final panel condenses high-dimensional text-encoded controls (specific-needs and per-status barrier dummies) into one row per group with the number of columns and pooled mean prevalence (the full per-column list is documented in Appendix EC.6.2).

Variable	Overall		Treated	
	Mean	Std	Mean	Std
<i>Child Demographics, Family, and Accommodation</i>				
age_6	0.240	0.427	0.235	0.424
age_7	0.100	0.299	0.101	0.301
age_8	0.044	0.206	0.044	0.206
age_9	0.038	0.192	0.038	0.192
age_10	0.081	0.273	0.086	0.280
age_11	0.100	0.300	0.106	0.308
age_12	0.069	0.254	0.071	0.257
age_13	0.061	0.239	0.060	0.238
age_14	0.078	0.268	0.077	0.267
age_15	0.087	0.282	0.085	0.279
age_16	0.073	0.260	0.069	0.254
age_17	0.028	0.166	0.027	0.161
is_female	0.474	0.499	0.477	0.499
is_syrian	0.970	0.171	0.970	0.169
is_single	0.995	0.067	0.996	0.065
turkish_low	0.116	0.321	0.119	0.324
turkish_medium	0.098	0.297	0.100	0.299
turkish_high	0.032	0.176	0.032	0.176

(continued)

Variable	Mean	Std	Mean	Std	
turkish_native	0.001	0.034	0.001	0.033	
had_turkey_school_experience	0.620	0.485	0.641	0.480	
family_two_parents	0.574	0.494	0.571	0.495	
family_single_mother	0.275	0.446	0.278	0.448	
family_single_father	0.107	0.309	0.108	0.311	
grandparent_present	0.066	0.249	0.065	0.247	
has_siblings	0.942	0.233	0.943	0.231	
is_eldest	0.342	0.474	0.340	0.474	
is_youngest	0.259	0.438	0.258	0.437	
sibling_treated_before	0.303	0.460	0.317	0.465	
household_size	6.048	3.005	6.074	2.999	
family_size	5.910	2.221	5.904	2.211	
real_income	10126.672	15745.793	10082.731	15677.236	104
real_rent	3980.682	7079.032	3972.211	7024.165	40
accom_formal_flat	0.850	0.357	0.850	0.357	
accom_formal_detached	0.092	0.289	0.094	0.292	
accom_formal_squatter	0.017	0.131	0.018	0.132	
accom_informal	0.010	0.102	0.008	0.090	
accom_formal_other	0.006	0.076	0.006	0.075	
shared_house	0.130	0.336	0.130	0.336	
<i>Risk-Assessment Summary (RiskLevel ref: 'low')</i>					
RiskLevel_medium	0.357	0.479	0.363	0.481	
RiskLevel_higher	0.176	0.381	0.158	0.365	
RiskLevel_highest	0.006	0.076	0.005	0.069	
is_miss_RiskLevel	0.085	0.279	0.083	0.275	
TypeOfCase_intermediate	0.705	0.456	0.724	0.447	
TypeOfCase_complex	0.172	0.377	0.154	0.361	
TypeOfCase_calling_assessment	0.034	0.180	0.034	0.182	
TypeOfCase_initial	0.005	0.072	0.005	0.071	
is_miss_TypeOfCase	0.085	0.279	0.083	0.275	
need_cat_alleged_perpetrator	0.000	0.009	0.000	0.009	
need_cat_child_at_risk	0.507	0.500	0.499	0.500	
need_cat_covid_19	0.000	0.016	0.000	0.016	
need_cat_disability	0.036	0.185	0.031	0.173	
need_cat_family_unity	0.001	0.036	0.001	0.034	
need_cat_gbv	0.004	0.062	0.003	0.055	
need_cat_parent_or_caregiver_with_specific_needs	0.000	0.012	0.000	0.012	
need_cat_persons_affected_by_disaster	0.086	0.281	0.081	0.273	
need_cat_serious_medical_condition	0.074	0.262	0.071	0.257	
need_cat_specific_legal_and_physical_protection_n	0.705	0.456	0.700	0.458	
need_cat_torture	0.002	0.042	0.002	0.042	
need_cat_unaccompanied_or_separated_child	0.008	0.090	0.007	0.082	
need_cat_victim_of_human_trafficking	0.000	0.011	0.000	0.009	
need_cat_woman_at_risk	0.000	0.011	0.000	0.009	
<i>Schooling-Barrier Summary (unified rollout)</i>					
any_barrier_reported	0.811	0.391	0.810	0.392	
n_barriers_unique	1.010	0.730	0.999	0.720	
barrier_BullyingPeer	0.022	0.146	0.021	0.144	
barrier_ChildLabour	0.051	0.219	0.047	0.212	
barrier_ChildRefusesSchool	0.089	0.284	0.083	0.276	
barrier_CulturalDiscrim	0.006	0.077	0.006	0.075	

(continued)

Variable	Mean	Std	Mean	Std
barrier_Disability	0.017	0.128	0.013	0.111
barrier_DocumentationIssues	0.030	0.169	0.020	0.140
barrier_EarthquakeOrCrisis	0.055	0.229	0.050	0.219
barrier_FinancialHardship	0.089	0.284	0.088	0.283
barrier_GenderForcedMarriage	0.008	0.087	0.007	0.083
barrier_HealthIssues	0.029	0.168	0.026	0.158
barrier_LanguageBarrier	0.018	0.133	0.018	0.134
barrier_NoFamilySupport	0.048	0.214	0.046	0.209
barrier_NotSchoolAge	0.145	0.352	0.141	0.348
barrier_RelocationDisplacement	0.015	0.121	0.013	0.114
barrier_SchoolAccessLocation	0.390	0.488	0.421	0.494
barrier_SeasonalAgriWork	0.000	0.000	0.000	0.000
<i>High-dim text-encoded controls (condensed; each group is a set of binary dummies)</i>				
Parental characteristics (presence-interacted dummies + ages) (30 cols)	2.070	—	2.067	—
max prevalence	30.720	—	—	—
need_sn_* (specific-needs detail) (27 cols)	0.060	—	0.057	—
max prevalence	0.697	—	—	—
Dropout/IrregAttend/NotAttend (per-status barrier detail) (37 cols)	0.038	—	0.037	—
max prevalence	0.282	—	—	—

EC.8. Pipeline Timing Correlations

The program’s annual rhythm is set by Türkiye’s academic calendar: the fall semester runs September through January, and most enrollment activity is concentrated in the months leading up to and immediately after the academic-year start. Figure EC.11 shows the monthly volume of first-contact education assessments and treatment deliveries across the eight cities for the 2020–2024 program years, demonstrating the same seasonal pattern in every year: a steady ramp-up from spring through late summer, a sharp peak in August–October at the fall registration window, and a tapering through the winter. The within-cohort first-come-first-served allocation rule discussed in subsection 3.3 is summarized below by Kendall’s τ between assessment date and treatment-delivery date for each city-year cohort.

Table EC.17: Kendall’s τ for Pipeline Stage Pairs by Cohort

Cohort	Assess $\rightarrow$ Action		Action $\rightarrow$ Enroll		Assess $\rightarrow$ Enroll		Enroll $\rightarrow$ Retention	
	Day	Week	Day	Week	Day	Week	Day	Week
adana_2021	0.884	0.886	0.661	0.672	0.597	0.606	0.270	0.258
adana_2022	0.925	0.928	0.709	0.718	0.671	0.679	0.004	-0.000
adana_2023	0.945	0.951	0.372	0.384	0.351	0.360	0.117	0.122
bursa_2021	0.616	0.637	0.449	0.479	0.380	0.401	0.186	0.122
bursa_2022	0.875	0.880	0.236	0.241	0.222	0.227	0.122	0.139
bursa_2023	0.895	0.903	0.354	0.362	0.319	0.328	0.254	0.265
gaziantep_2021	0.615	0.632	0.211	0.212	0.159	0.162	0.132	0.138
gaziantep_2022	0.946	0.950	0.170	0.179	0.183	0.190	0.183	0.200

(continued)

	Day	Week	Day	Week	Day	Week	Day	Week
gaziantep_2023	0.855	0.864	0.159	0.173	0.175	0.185	0.151	0.163
istanbul_2021	0.714	0.729	0.271	0.277	0.229	0.233	0.346	0.344
istanbul_2022	0.893	0.900	0.413	0.433	0.391	0.405	-0.105	-0.090
istanbul_2023	0.804	0.811	0.316	0.326	0.254	0.263	0.189	0.201
izmir_2021	0.667	0.673	0.555	0.576	0.434	0.446	-0.061	-0.050
izmir_2022	0.913	0.916	0.559	0.571	0.511	0.522	0.147	0.152
izmir_2023	0.857	0.858	0.344	0.346	0.193	0.189	0.133	0.135
konya_2021	0.702	0.714	0.257	0.276	0.234	0.246	0.095	0.105
konya_2022	0.948	0.953	0.425	0.438	0.423	0.435	0.042	0.046
konya_2023	0.931	0.937	0.234	0.236	0.233	0.236	0.164	0.235
mersin_2021	0.757	0.771	0.411	0.430	0.346	0.356	0.244	0.225
mersin_2022	0.893	0.899	0.332	0.343	0.304	0.312	0.131	0.144
mersin_2023	0.911	0.913	0.411	0.419	0.379	0.387	0.160	0.169
sanliurfa_2021	0.729	0.742	0.333	0.341	0.297	0.305	0.266	0.234
sanliurfa_2022	0.900	0.909	0.405	0.432	0.401	0.424	0.011	-0.008
sanliurfa_2023	0.867	0.870	0.143	0.154	0.156	0.167	0.094	0.096
<i>Min</i>	0.615	0.632	0.143	0.154	0.156	0.162	-0.105	-0.090
<i>Mean</i>	0.835	0.843	0.364	0.376	0.327	0.336	0.136	0.139
<i>Median</i>	0.879	0.883	0.349	0.354	0.311	0.320	0.140	0.141
<i>Max</i>	0.948	0.953	0.709	0.718	0.671	0.679	0.346	0.344

Acknowledgments

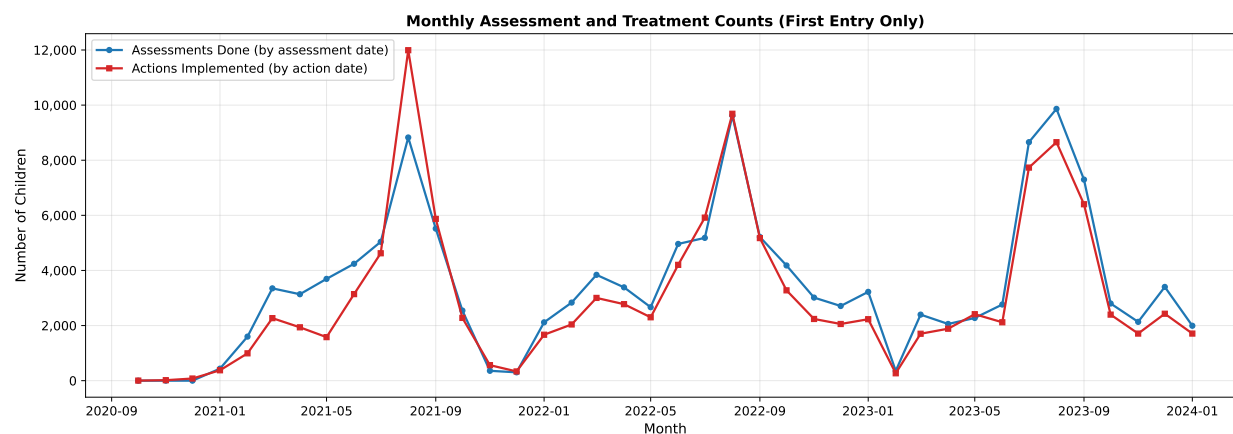


Figure EC.11 Monthly volume of first-contact education assessments and treatment deliveries, 2020–2024, pooled across the eight cities in the analysis sample. Each program year shows the same seasonal pattern: a steady ramp through spring and summer, a sharp peak in August–October at the fall enrollment window, and a tapering through winter as the cohort closes. The COVID-19 disruption in 2020 and the February 2023 earthquake (visible as the dip in 2023 Q1 for the southeastern cities) are the two largest deviations from the recurring pattern.